

Abdul Wahid

CND Lab Report (Volume 2 of 3)

EC-Council CND (Certified Network Defender)

Data Security – Transit Encryption, Secure Communication, Backup & Recovery

Executive Summary

This document covers four practical labs focused on data protection and recovery.

It begins with enabling SSL/TLS encryption for a locally hosted WordPress site to secure data in transit. It then demonstrates secure email communication using OpenPGP (Gpg4win & Kleopatra) through key pair creation, encryption, and decryption. Next, it shows how to create reliable file backups using AOMEI Backupper Standard. Finally, it covers recovering intentionally deleted files using EaseUS Data Recovery Wizard. Together, these labs provide hands-on experience in encrypting data in transit, protecting communications, backing up critical files, and recovering lost data.

Table of Content

Encrypting Data in Transit Using SSL

- Generate self-signed SSL certificate (key, CSR, CRT)
- Configure Apache (enable ssl_module + edit httpd-ssl.conf)
- Access WordPress site over HTTPS (https://localhost/cnd)

Secure Email Communication Using PGP

- Install Gpg4win + Kleopatra
- Create OpenPGP key pair (public + private)
- Export public key & backup private key
- Encrypt a message → Decrypt it successfully

Performing Data Backup Using AOMEI Backupper

- Install AOMEI Backupper Standard
- Create test folder + sample file
- Perform full file backup to a separate destination folder
- Verify backup is created successfully

File Recovery Using EaseUS Data Recovery Wizard

- Install EaseUS Data Recovery Wizard
- Intentionally delete the Lab 6 backup folder
- Scan Local Disk C: and recover the deleted files
- Verify recovered files open correctly

Encrypting Data in Transit Using SSL

Machine: Server-1 (Windows Server 2019, running WampServer + WordPress)

Summary: SSL/TLS encrypts data flowing between a web browser and a web server, so data can't be intercepted/read in transit. This lab enables SSL on a locally hosted WordPress site.

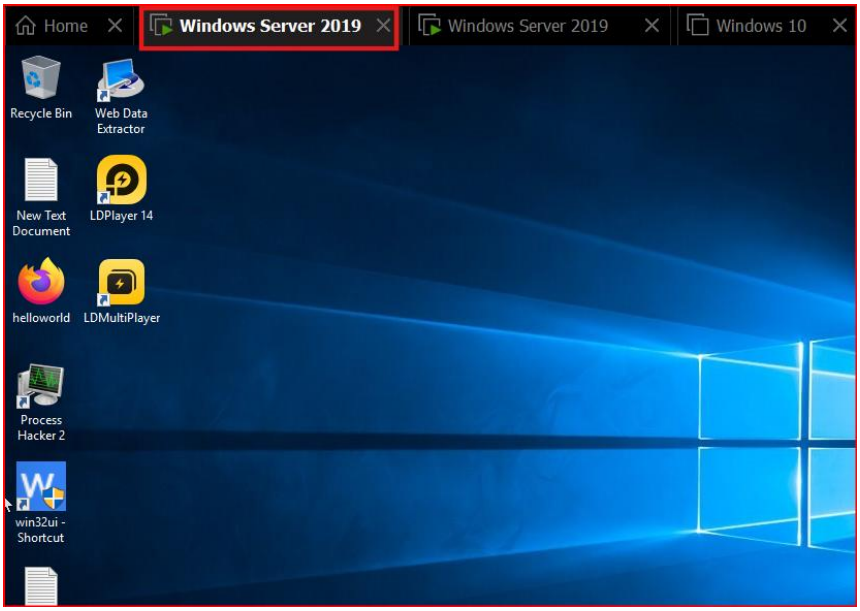
Objective: Generate an SSL certificate, configure Apache (inside WampServer) to use it, and access the site over HTTPS.

Prerequisite: WampServer must be running with a WordPress site already set up at C:\wamp64\www\cnd (accessible via http://localhost/cnd). If you haven't set this up yet, tell me and I'll walk you through installing WampServer + WordPress first.

Key Concepts

- Generated a self-signed SSL certificate (server.key/pem/crt)
- Configured Apache to serve over port 443
- Fixed several real config issues along the way (missing #, wrong file paths, wrong key file, wrong DocumentRoot) — solid troubleshooting work
- Confirmed the WordPress "cnd" site loads over HTTPS with an encrypted channel (the "Not Secure" label is just because it's self-signed, not a failure)

Launch **Server-1**



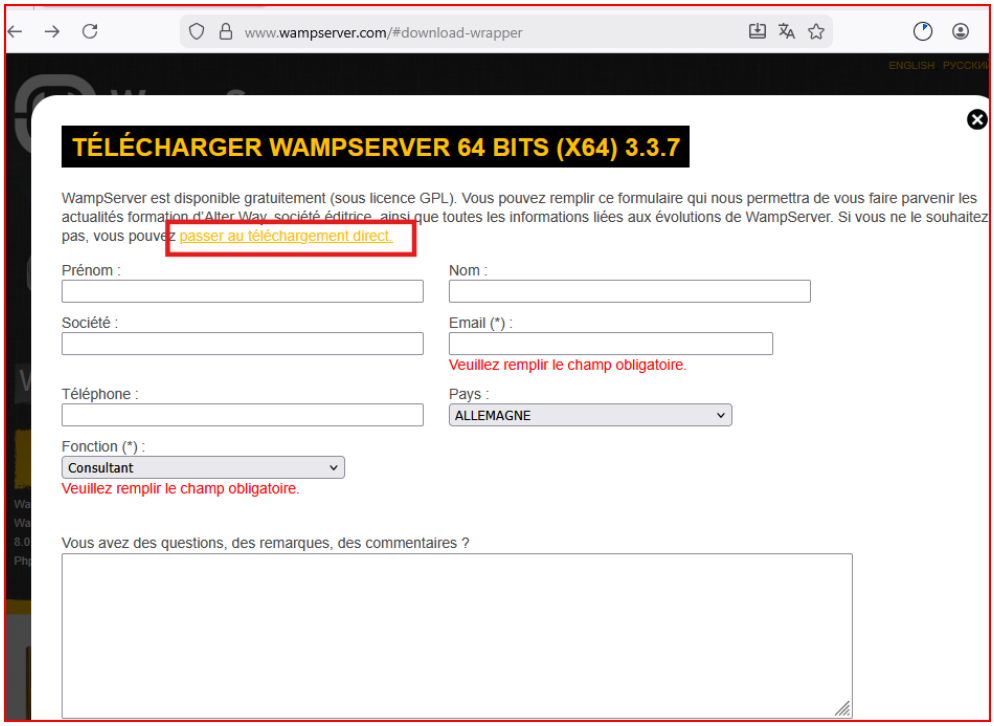
Wamp server Installation

On **Server-1**, open a browser, go to **wampserver.com**, download WampServer (64-bit, matching your PHP version — pick the default one offered).





Click the selected box and your installation will start.



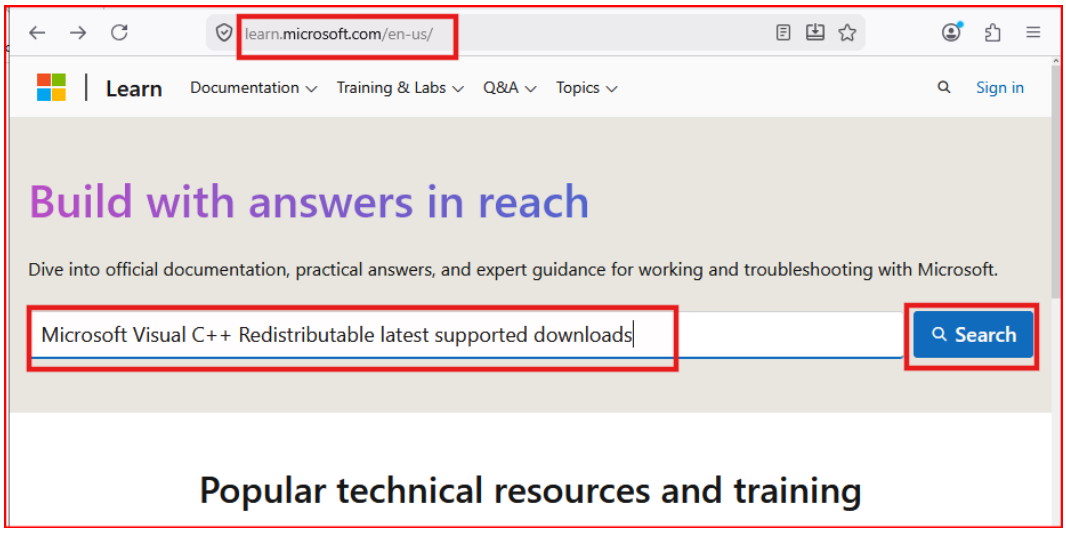
Run the installer. It'll likely ask you to install **Visual C++ Redistributables** first if missing — install those if prompted.

Download Visual C++ Redistributable (if missing)

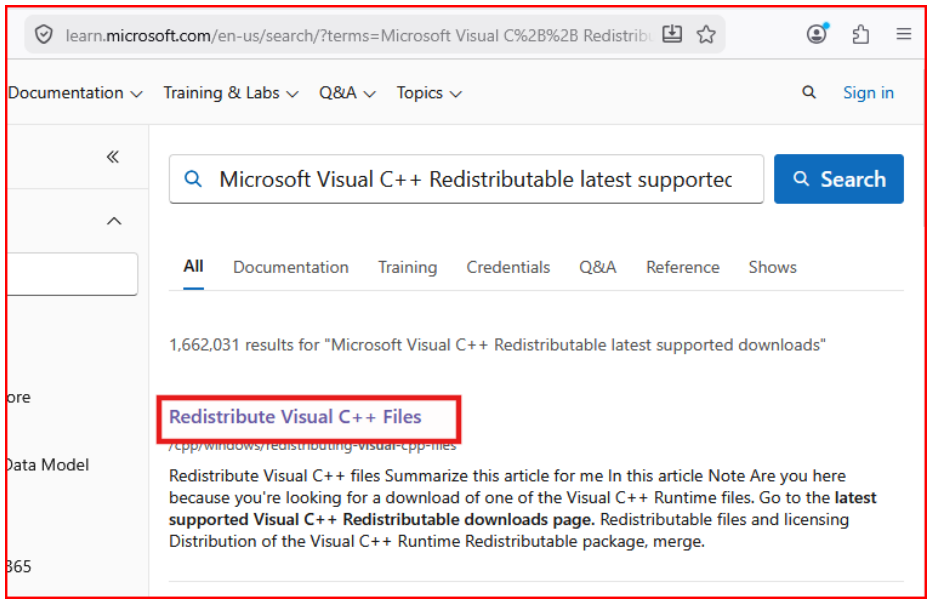
Your Windows 10 is **64-bit**, so it can run **64-bit applications**.

But WampServer includes or supports **both 64-bit and 32-bit (x86) components**. Some PHP extensions, Apache modules, or tools may be compiled as 32-bit.

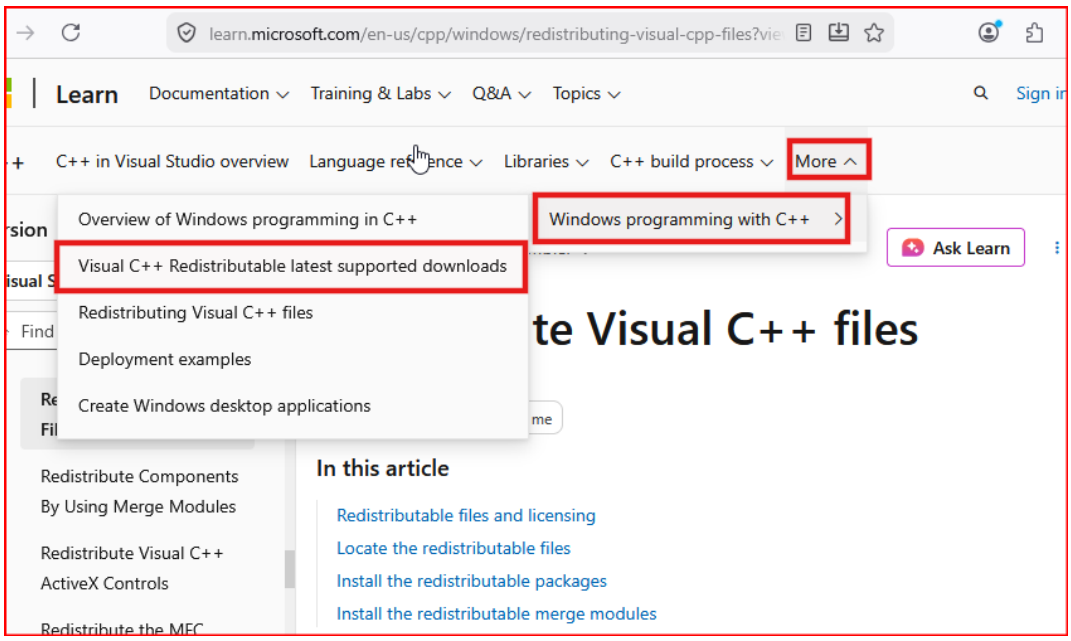
Because of this, the WampServer installer **checks for both x86 and x64 Visual C++ runtimes**.



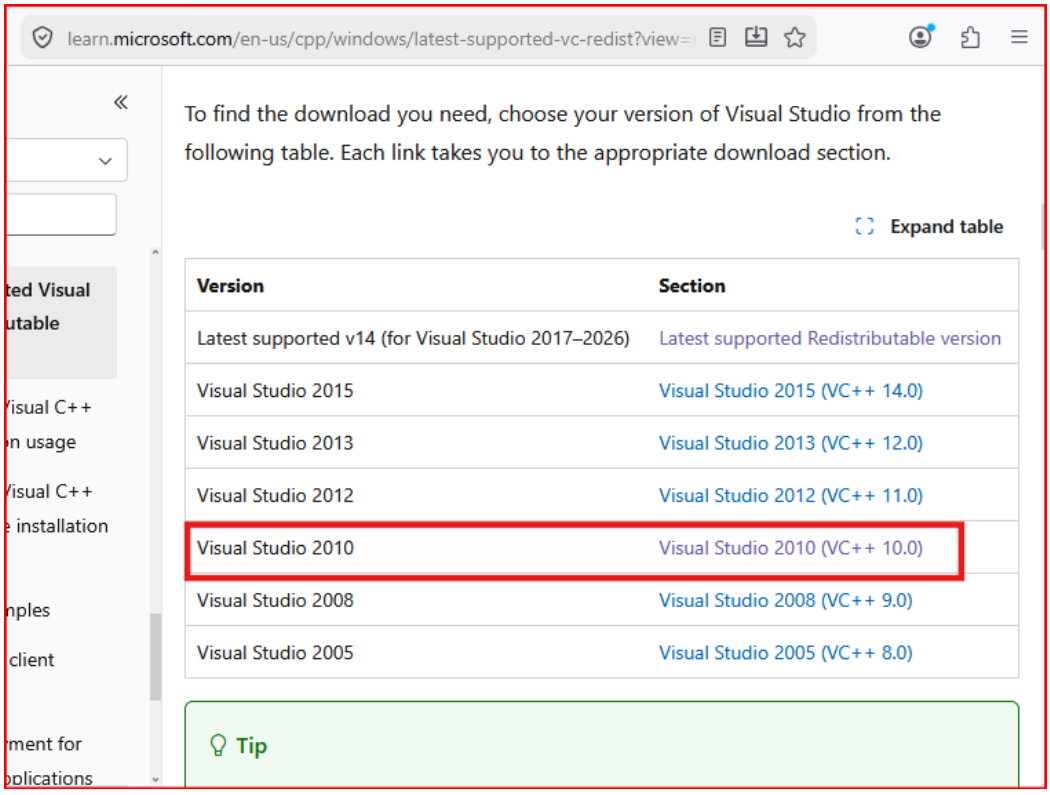
Click on Redistribute Visual C++ Files



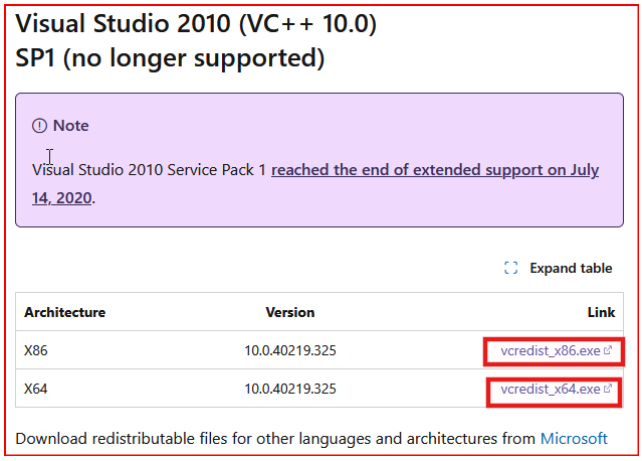
Click on More-> Windows Programming with C++ -> Visual C++Redistributable latest supported downloads



Select Visual Studio 2010



Download both files & run them to install them



Visual Studio 2010 (VC++ 10.0)
SP1 (no longer supported)

Note

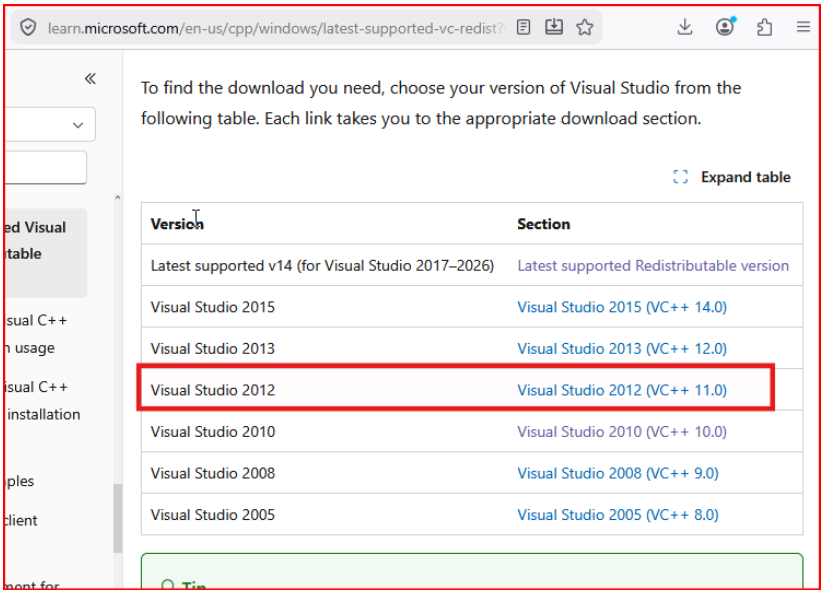
Visual Studio 2010 Service Pack 1 reached the end of extended support on July 14, 2020.

Expand table

Architecture	Version	Link
X86	10.0.40219.325	vcredist_x86.exe
X64	10.0.40219.325	vcredist_x64.exe

Download redistributable files for other languages and architectures from Microsoft

Select Visual Studio 2011



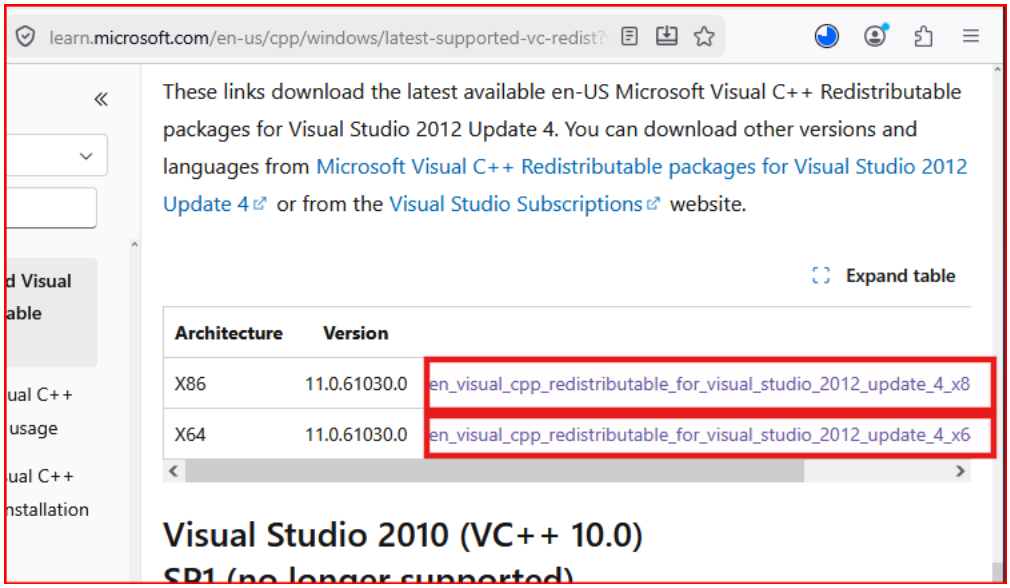
learn.microsoft.com/en-us/cpp/windows/latest-supported-vc-redist?

To find the download you need, choose your version of Visual Studio from the following table. Each link takes you to the appropriate download section.

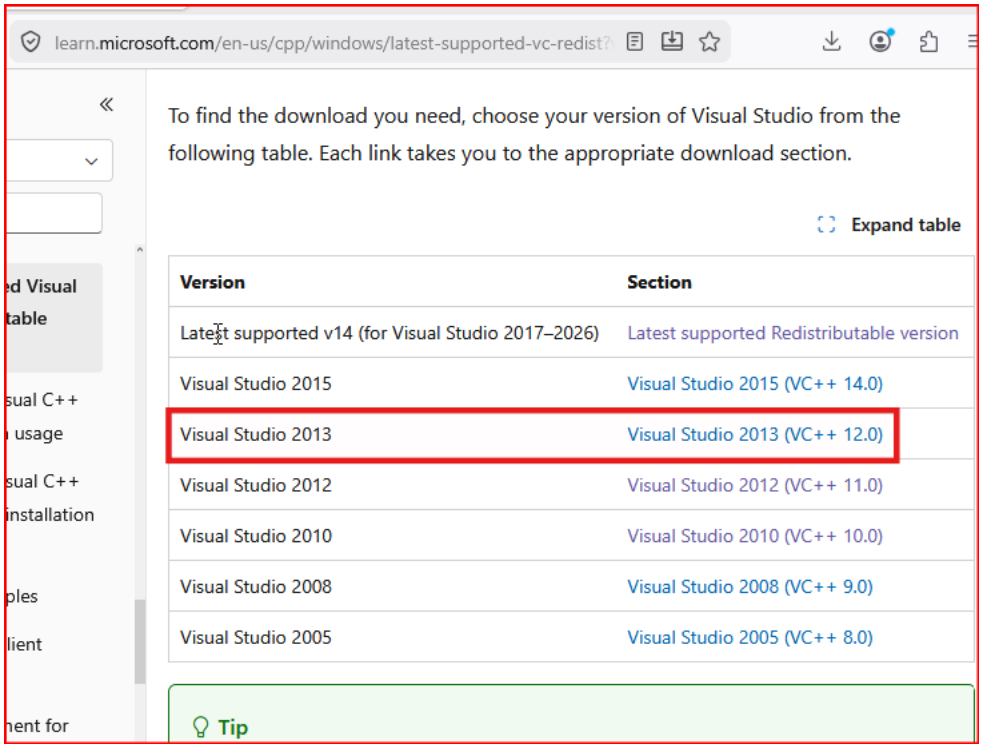
Expand table

Version	Section
Latest supported v14 (for Visual Studio 2017–2026)	Latest supported Redistributable version
Visual Studio 2015	Visual Studio 2015 (VC++ 14.0)
Visual Studio 2013	Visual Studio 2013 (VC++ 12.0)
Visual Studio 2012	Visual Studio 2012 (VC++ 11.0)
Visual Studio 2010	Visual Studio 2010 (VC++ 10.0)
Visual Studio 2008	Visual Studio 2008 (VC++ 9.0)
Visual Studio 2005	Visual Studio 2005 (VC++ 8.0)

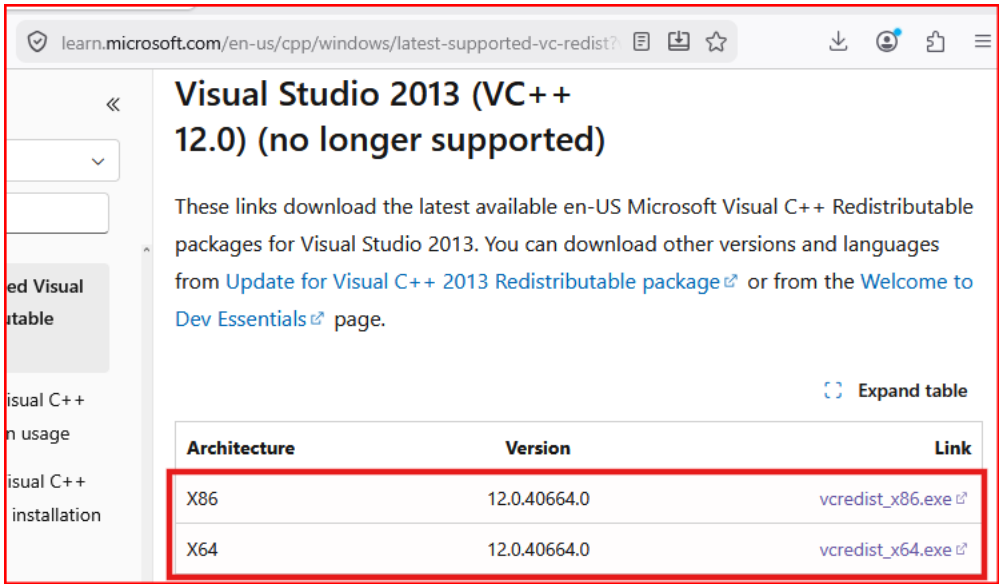
Download both files and run them to install them



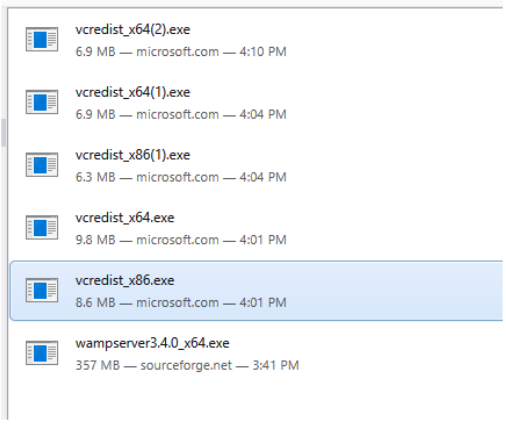
Select the Visual Studio 2012



Download both files and run them to install them



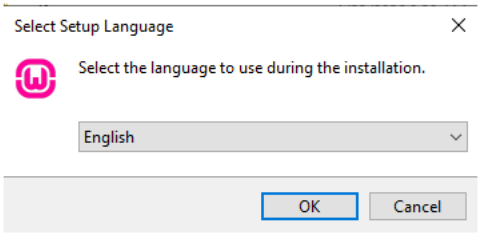
At the end all these 6 files should be installed



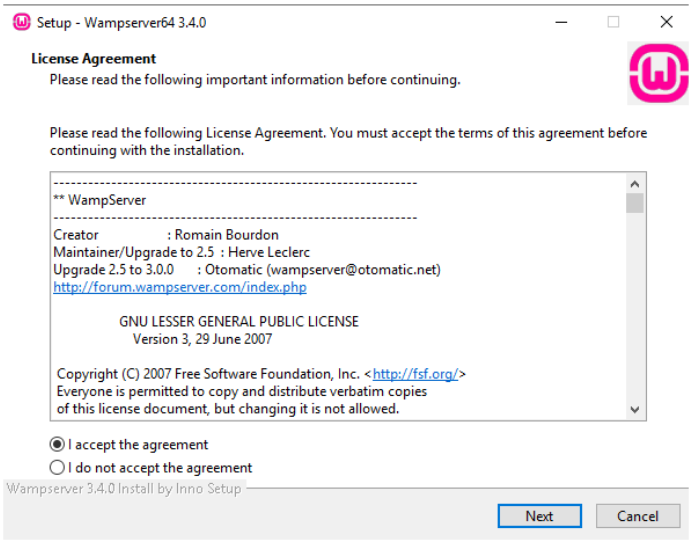
Now restart your system

Setting up Wamp Server

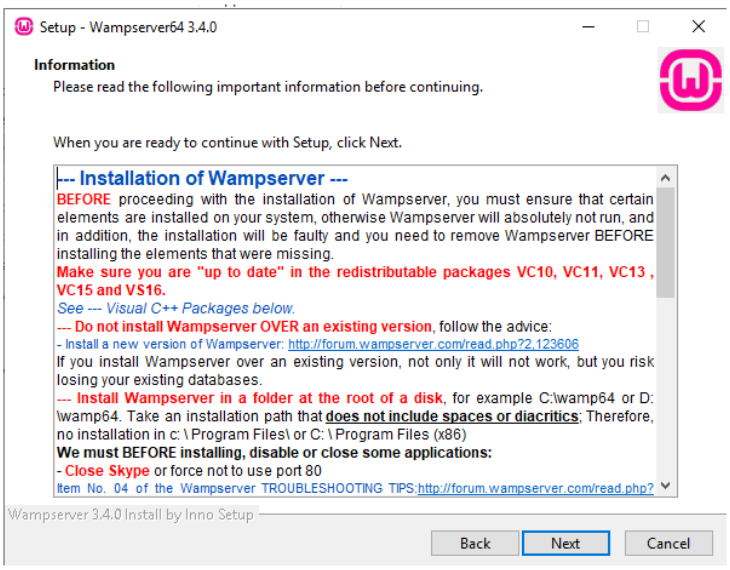
Complete the installation, leave "Launch WampServer" checked at the end.



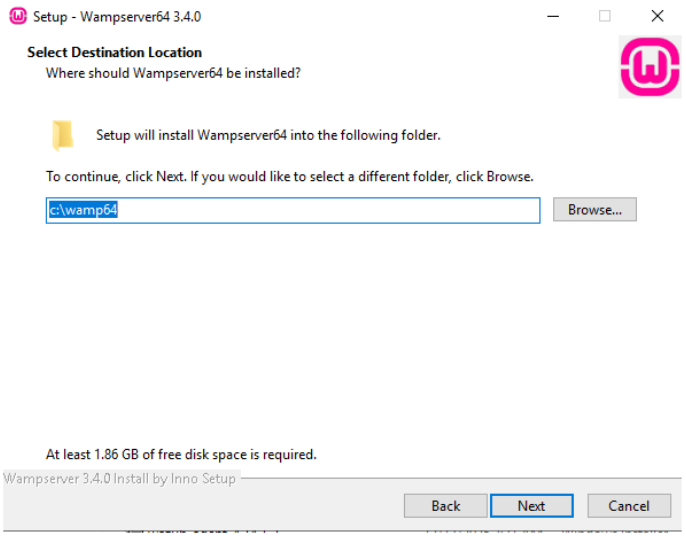
Select the acceptance agreement and click Next



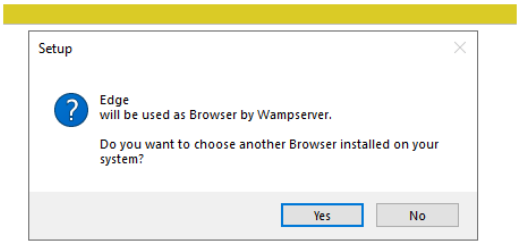
Click Next



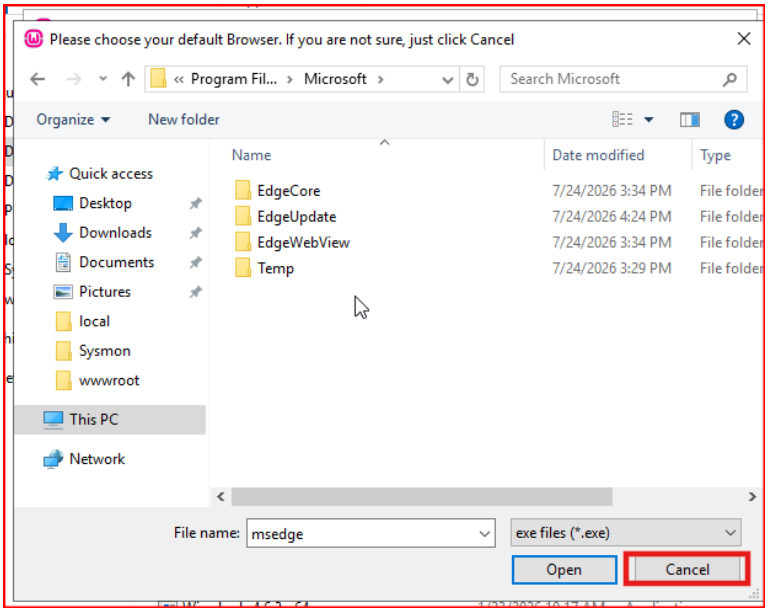
Browse the place to install the server



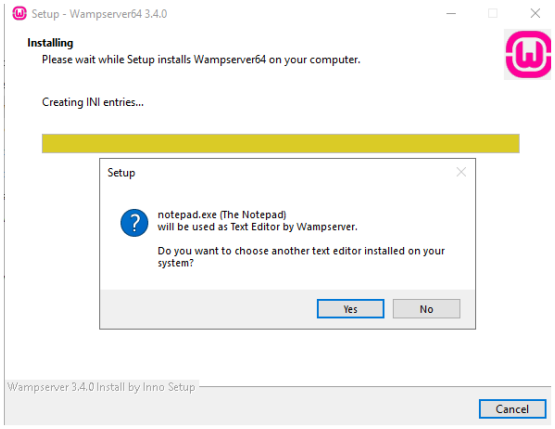
Click Yes



Select Cancel



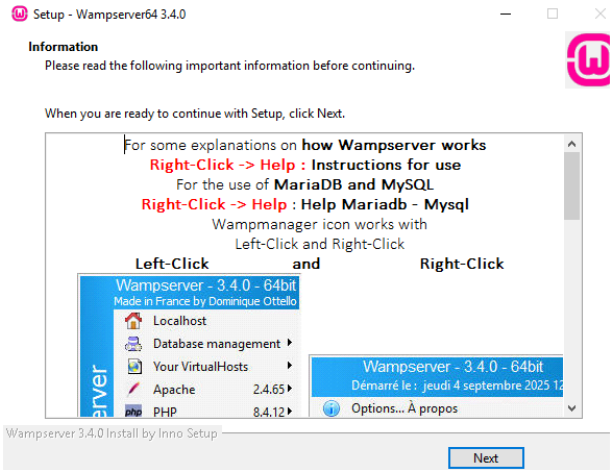
Click yes



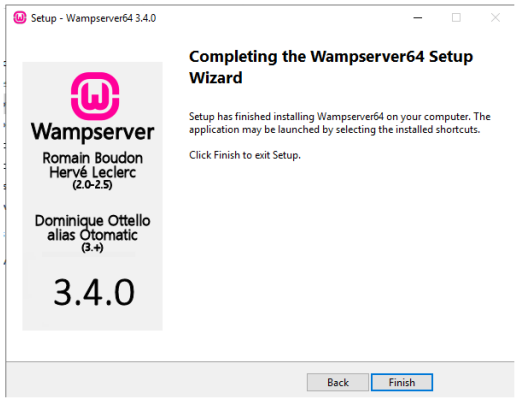
Then it will ask you for notepad so in search bar

Search = notepad.exe -> Press Enter

Click Next



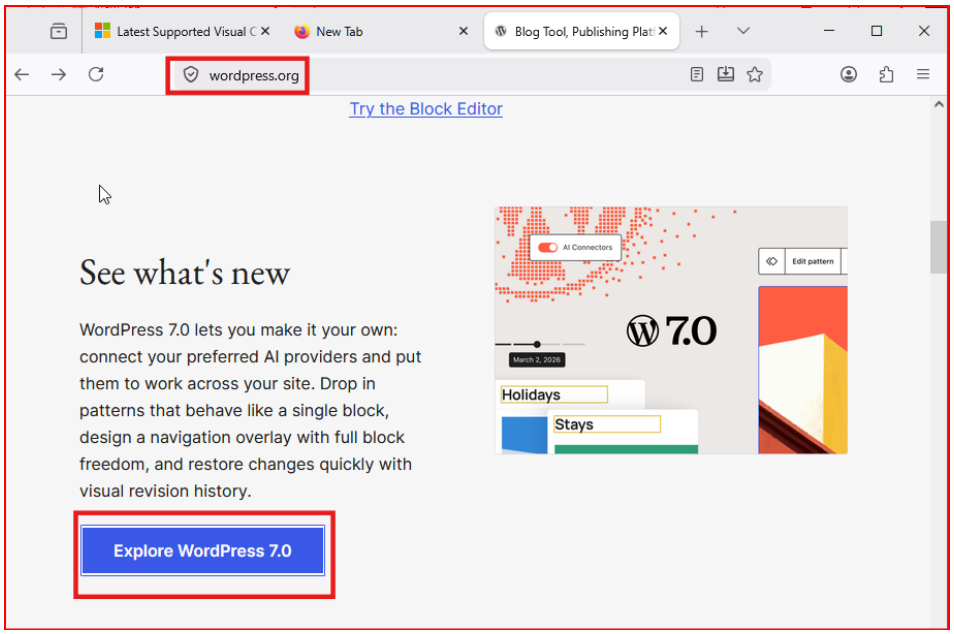
Select Finish



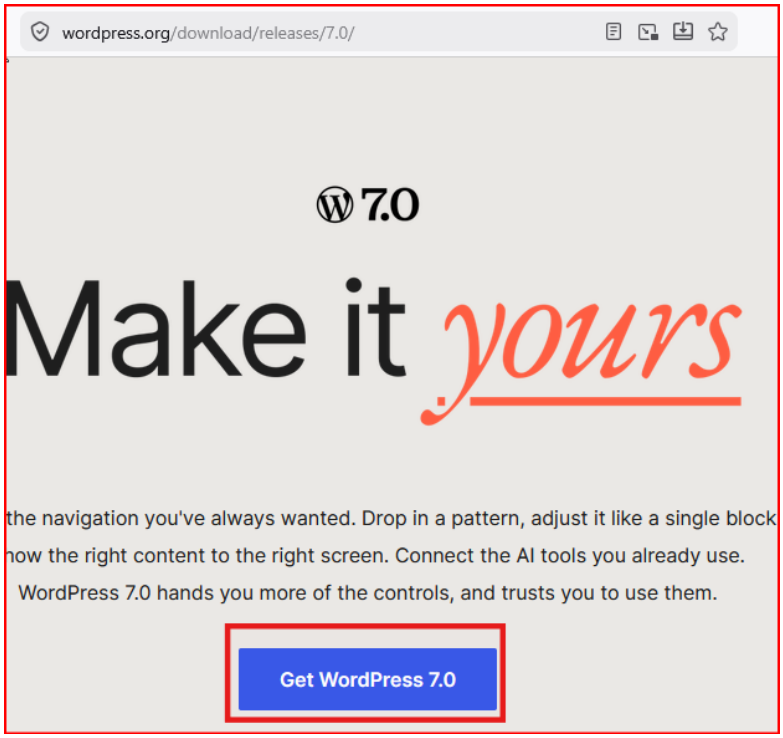
Download WordPress

URL: wordpress.org/download

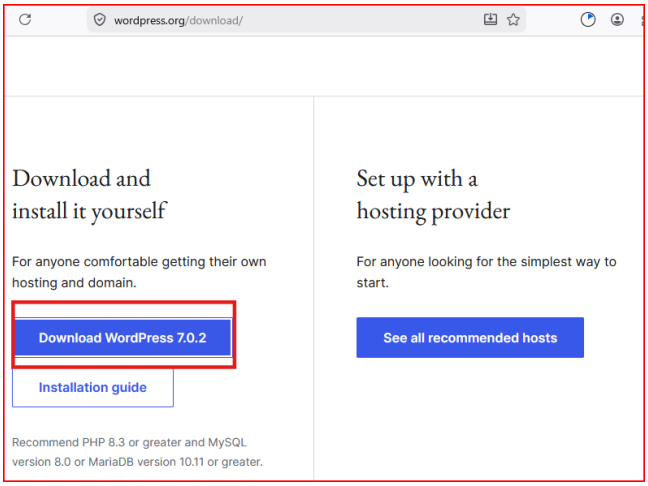
Click Explore



Select Get WordPres 7.0

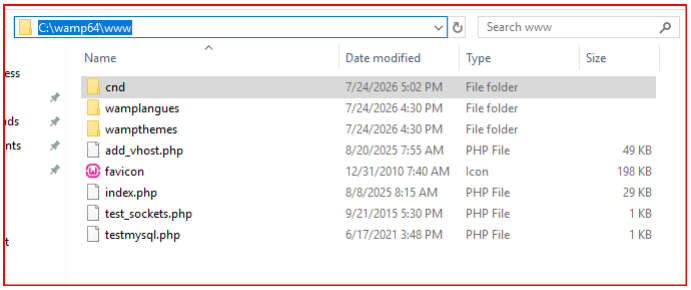


Download the wordpress 7.0.2



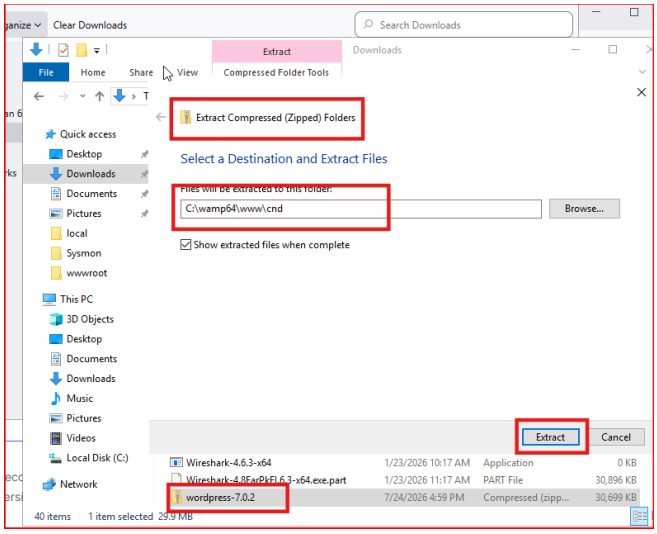
Navigate to C:\wamp64\www\.

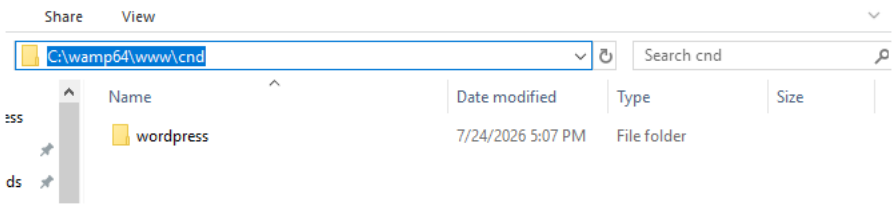
Create a new folder here named **cnd**.



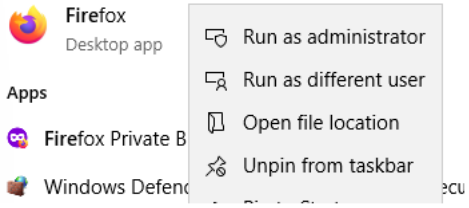
Extract the zip — you'll get a folder named wordpress.

Copy all contents from the extracted wordpress folder into C:\wamp64\www\cnd.

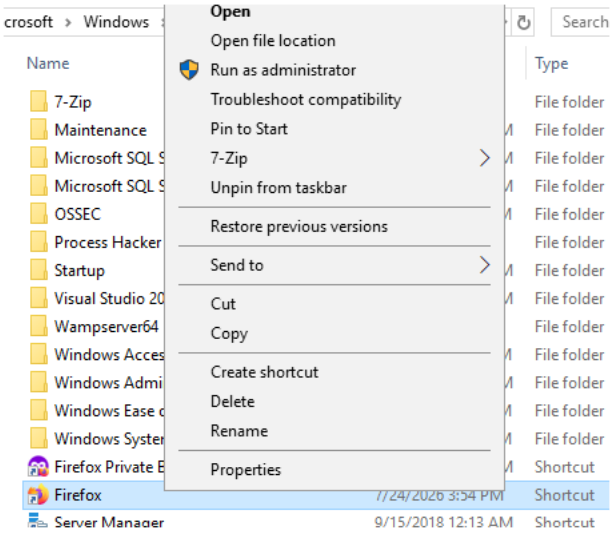




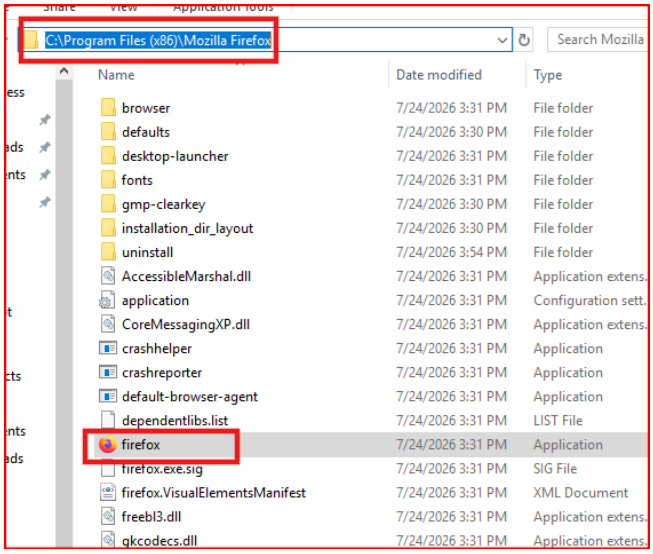
Now search firefox and right click on it select Open file location



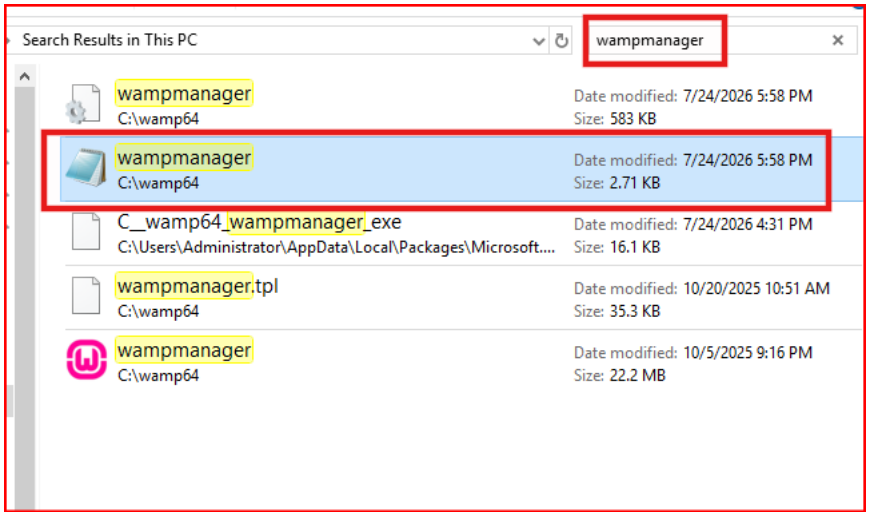
Now again select firefox and right click on it select Open file location



Remember the path of firefox

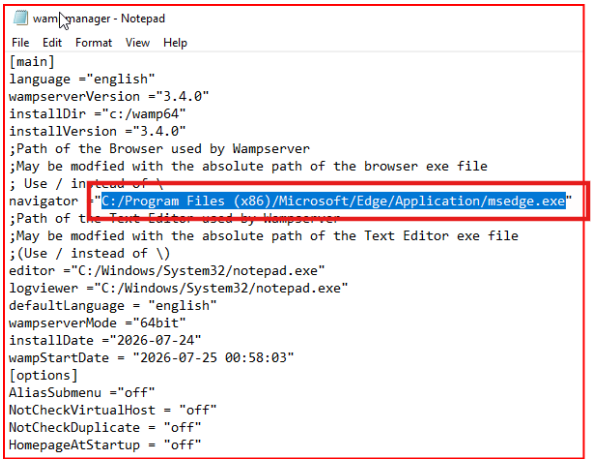


Now open file explorer and Search for "wampmanager" and open the notepad file of wampmanager



Now replace the file location with the firefox file location

Before -> After

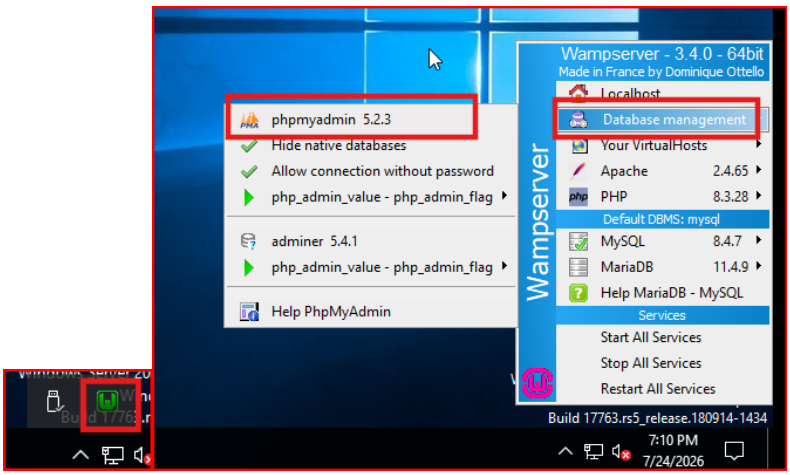




Click the WampServer tray icon → **Database management** → **phpMyAdmin** (opens in browser).



Green shows working properly

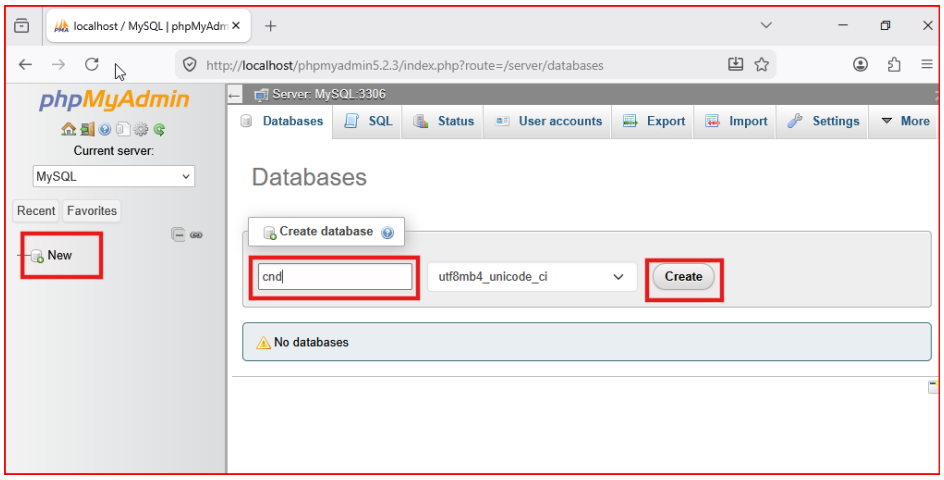


WordPress Setting up (New Database)

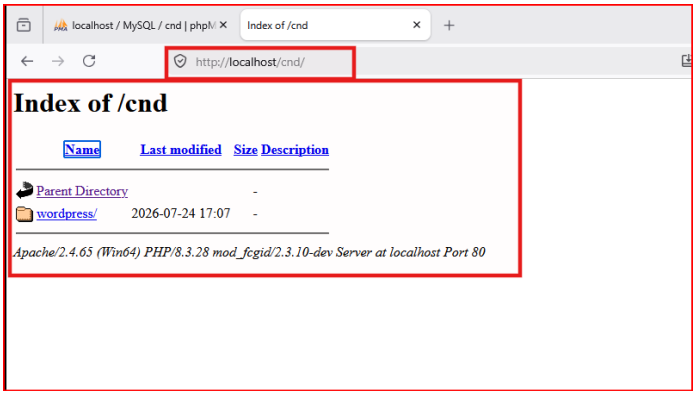
Log in (default: username root, no password, unless you set one).



Click **New** (left sidebar) → create a database named cnd → click Create.

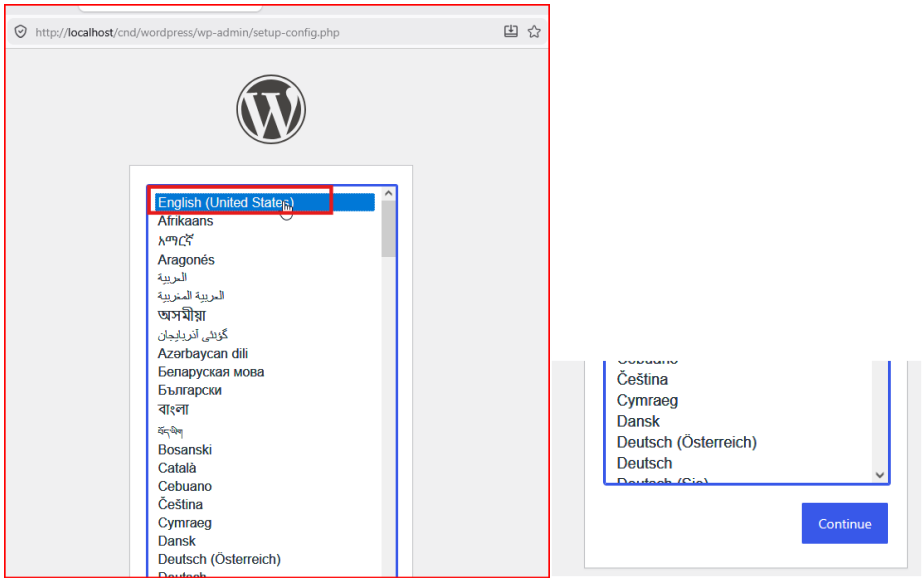


Open browser, go to http://localhost/cnd.

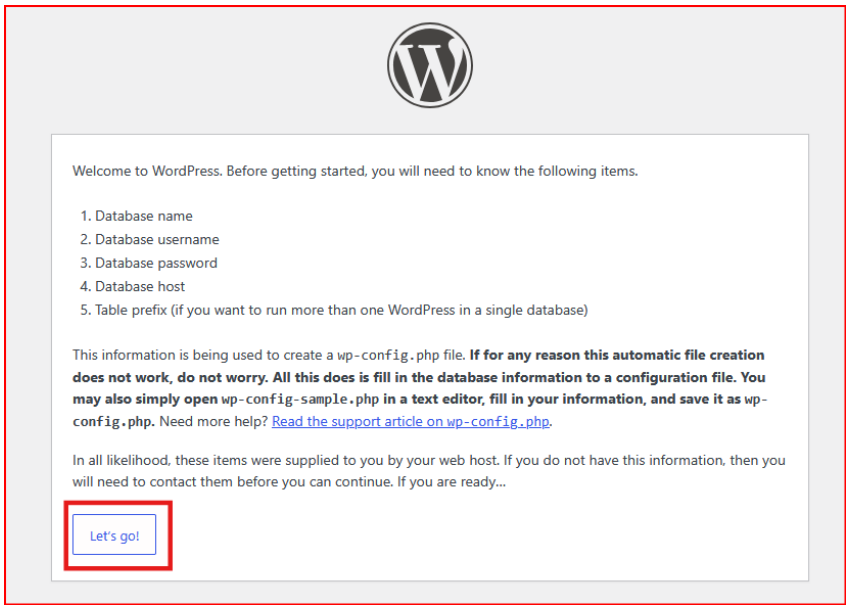


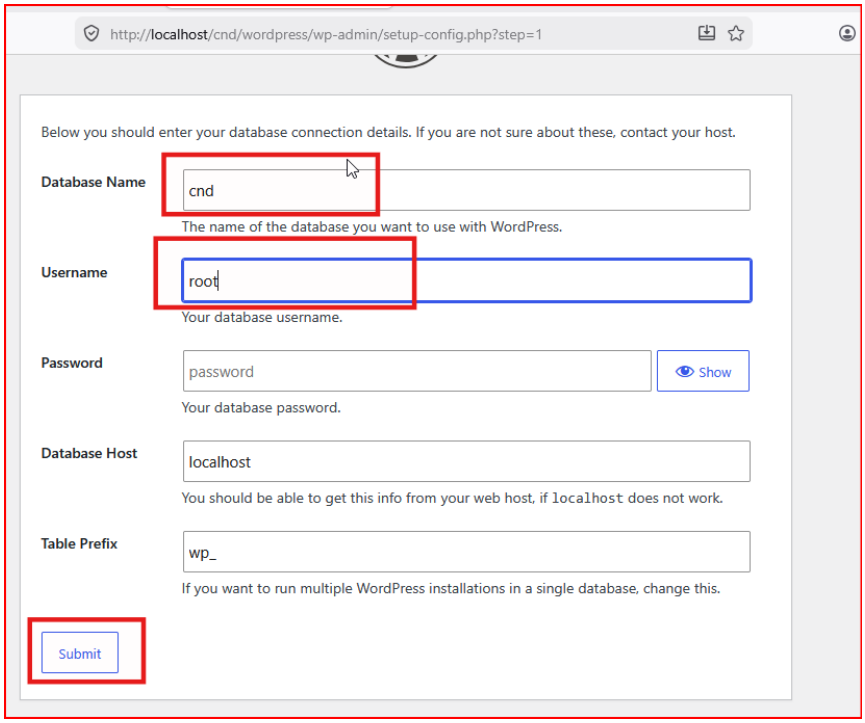


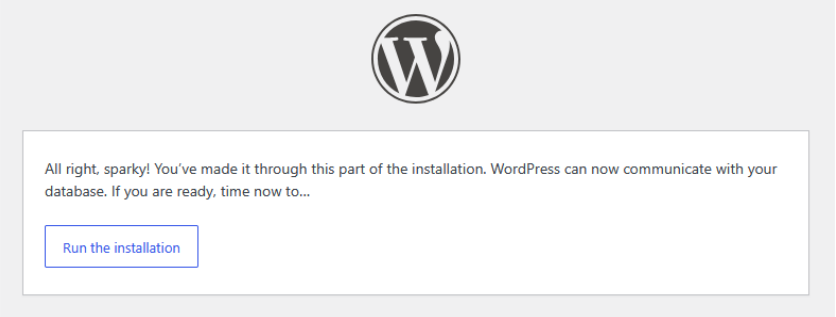
WordPress setup wizard should appear — select language → Continue.



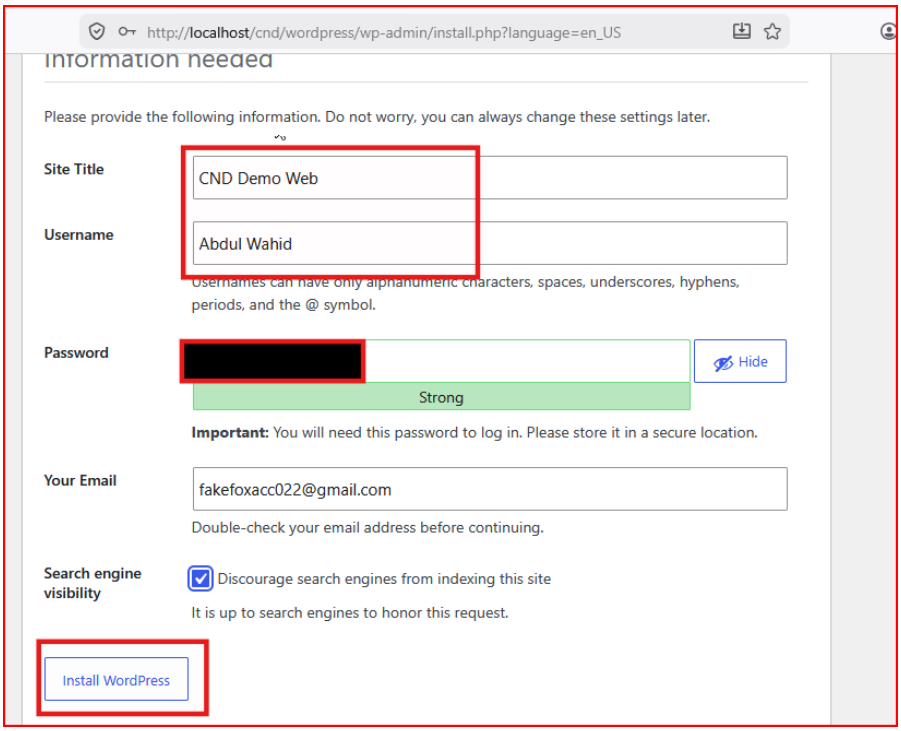
Enter database details: Database Name: cnd, Username: root, Password: (leave blank unless you set one), Database Host: localhost, Table Prefix: wp_ → Submit → Run the install.



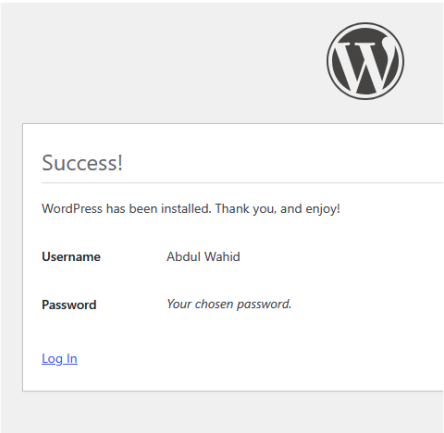


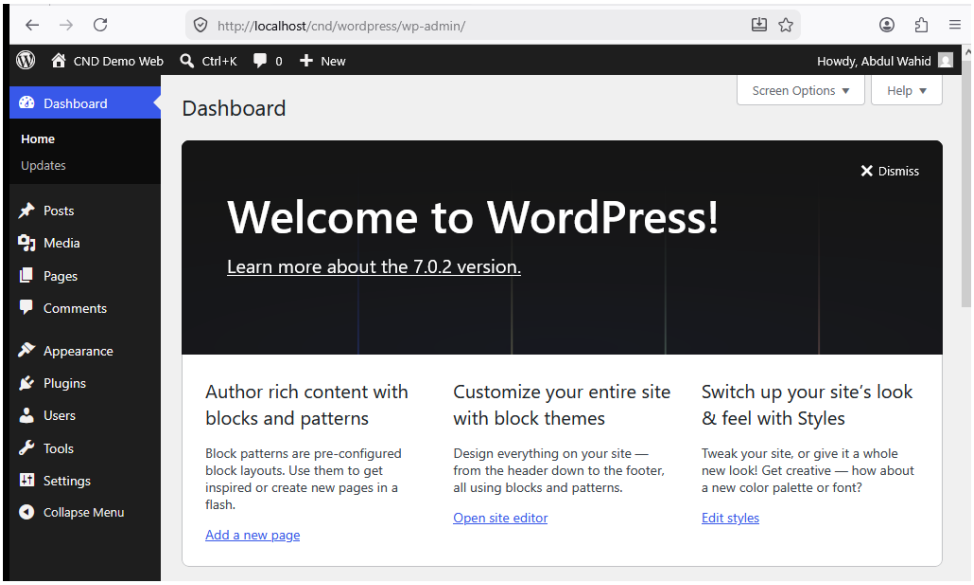


Fill in Site Title (e.g., "CND Demo Website"), create an admin username/password, enter your email → Click **Install WordPress**.

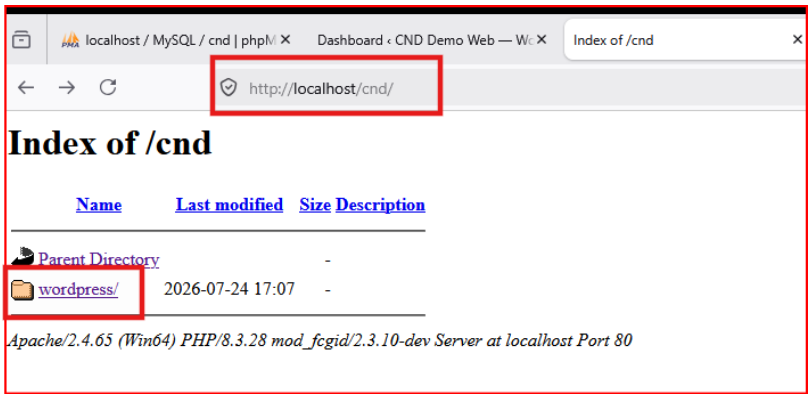


Log in with the admin credentials you just created.

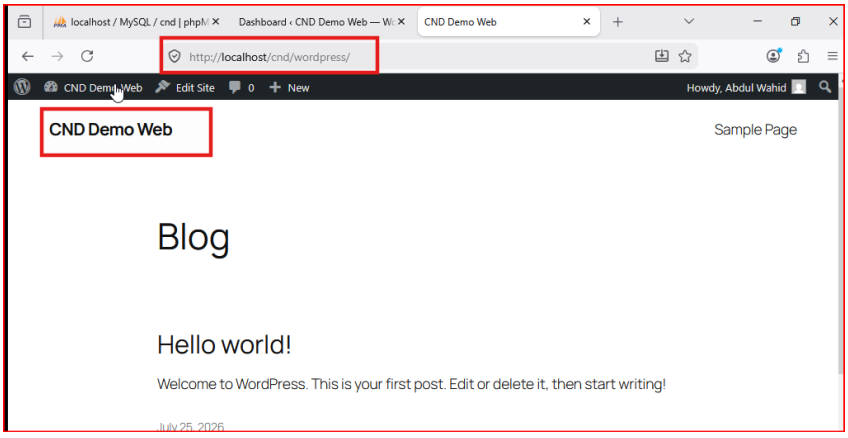




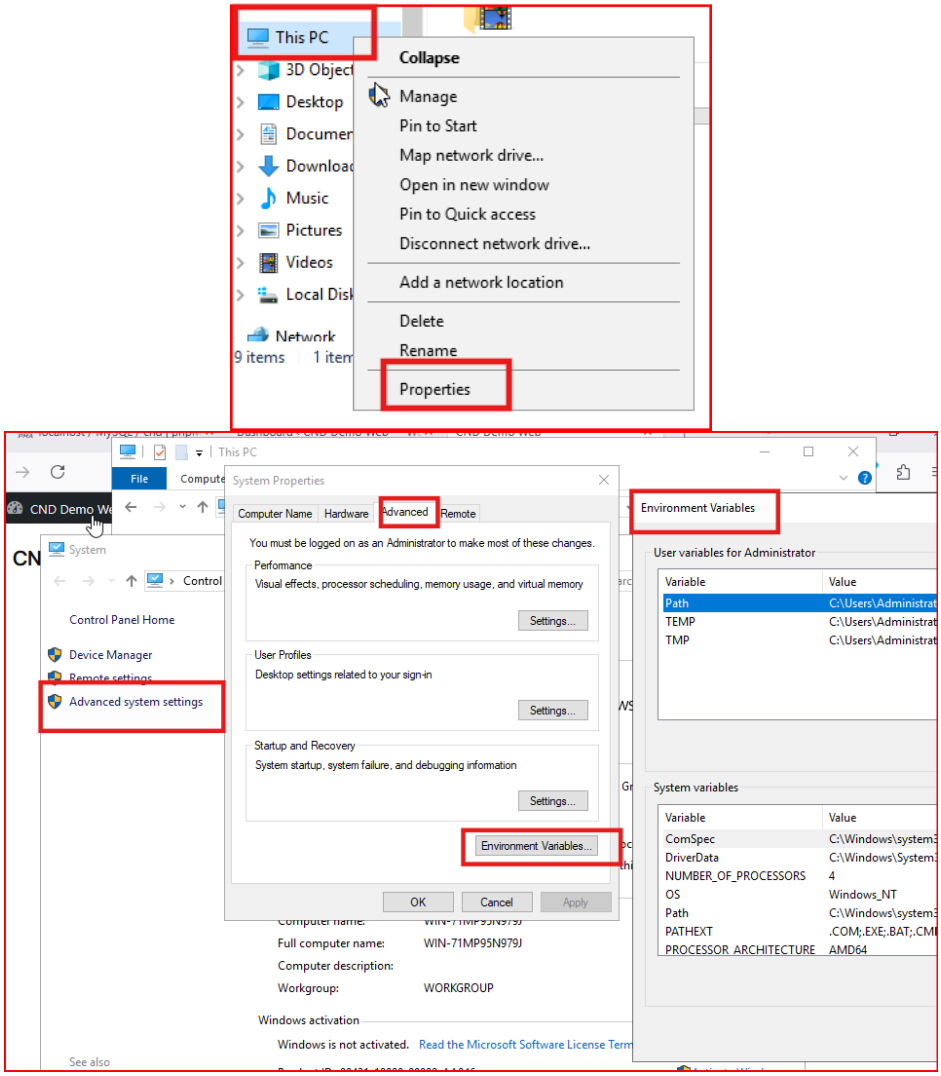
Open a new tab, go to http://localhost/cnd — you should see the WordPress site live.



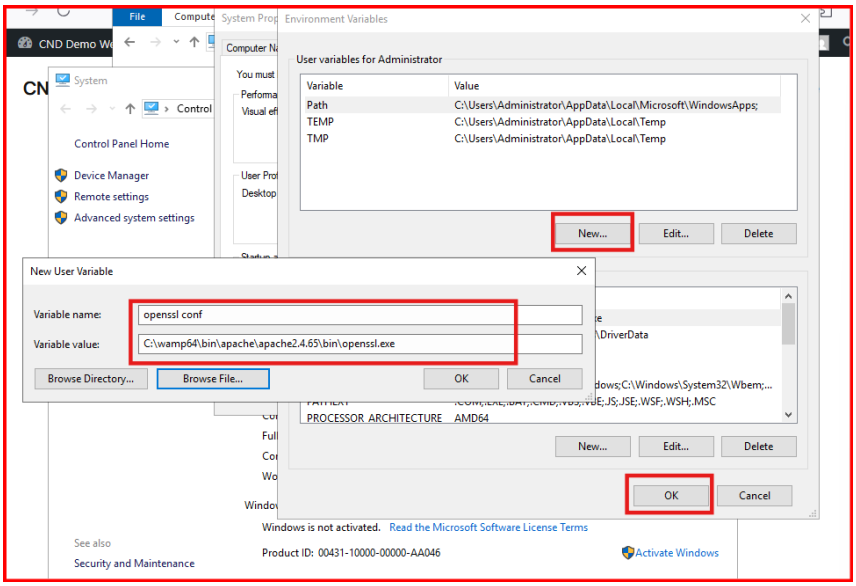
Confirm http://localhost/cnd loads in browser (plain HTTP, not HTTPS yet) — this is your baseline before enabling SSL.

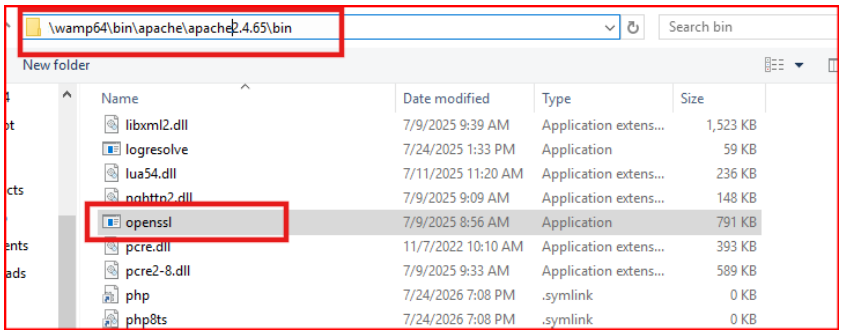


Open **File Explorer** → right-click **This PC** → **Properties** → **Advanced system settings** → **Advanced** tab → **Environment Variables**.

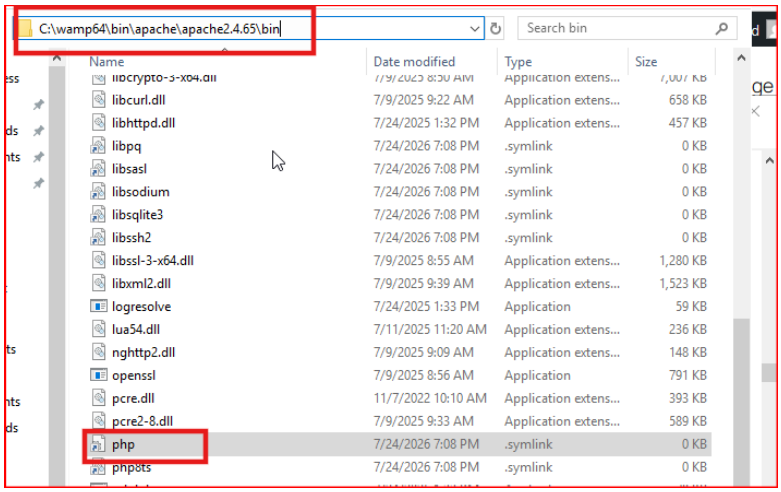


Under "User variables," click **New**. Variable name: openssl_conf. Variable value: C:\wamp64\bin\apache\apache2.4.58\bin\apache2.4.58\conf\openssl.cnf (adjust apache version number to match yours). Click OK, OK, OK to close all windows.

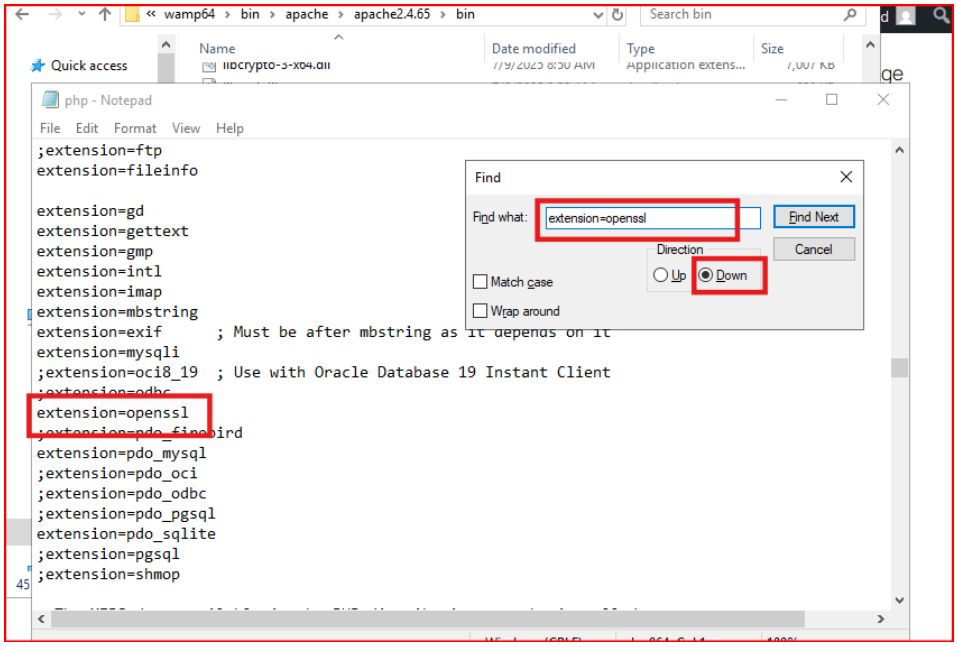




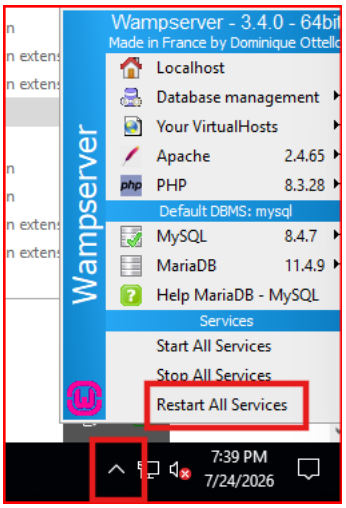
Navigate to C:\wamp64\bin\apache\apache2.4.58\bin, open **php.ini** with Notepad++.



Find the line ;extension=openssl — remove the ; to uncomment it. Save the file.

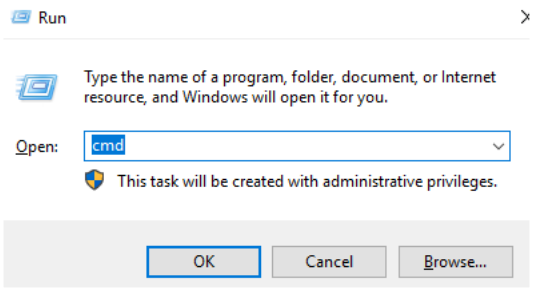


Click WampServer tray icon → **Restart All Services**. Wait for green icon.



Complete SSL Certificate Commands for WampServer

open Command Prompt as Administrator.



Step 1 — Open the Apache bin folder

cd /d C:\wamp64\bin\apache\apache2.4.65\bin

This moves Command Prompt to the folder containing openssl.exe.

```
C:\Users\Administrator>cd /d C:\wamp64\bin\apache\apache2.4.65\bin
```

Your prompt should now show:

C:\wamp64\bin\apache\apache2.4.65\bin>

Step 2 — Set the correct OpenSSL configuration file

set "OPENSSL_CONF=C:\wamp64\bin\php\php8.3.28\extras\ssl\openssl.cnf"

```
C:\wamp64\bin\apache\apache2.4.65\bin>set "OPENSSL_CONF=C:\wamp64\bin\php\php8.3.28\extras\ssl\openssl.cnf"
```

This tells OpenSSL where its configuration file is located.

Step 3 — Verify the configuration path

echo %OPENSSL_CONF%

It should display:

C:\wamp64\bin\php\php8.3.28\extras\ssl\openssl.cnf

```
C:\wamp64\bin\apache\apache2.4.65\bin>echo %OPENSSL_CONF%
C:\wamp64\bin\php\php8.3.28\extras\ssl\openssl.cnf
```

Step 4 — Confirm that the configuration file exists

if exist "%OPENSSL_CONF%" (echo CONFIG FILE FOUND) else (echo CONFIG FILE NOT FOUND)

The expected result is:

```
C:\wamp64\bin\apache\apache2.4.65\bin>if exist "%OPENSSL_CONF%" (echo CONFIG FILE FOUND) else (echo CONFIG FILE NOT FOUND)
CONFIG FILE FOUND
```

CONFIG FILE FOUND

Do not continue if it says CONFIG FILE NOT FOUND.

Step 5 — Check that OpenSSL is working

openssl version

```
C:\wamp64\bin\apache\apache2.4.65\bin>openssl version
OpenSSL 3.5.1 1 Jul 2025 (Library: OpenSSL 3.5.1 1 Jul 2025)
```

This should display the installed OpenSSL version.

Step 6 — Remove previous certificate files

del /Q server.key server.pem server.csr server.crt localhost.ext 2>nul

```
C:\wamp64\bin\apache\apache2.4.65\bin>del /Q server.key server.pem server.csr server.crt localhost.ext 2>nul
```

This removes incomplete certificate files from previous attempts.

Step 7 — Generate an encrypted private key

openssl genrsa -des3 -out server.key 2048

This creates a password-protected 2048-bit private key.

When asked:

Enter PEM pass phrase:

Enter a password, for example: Admin123

Enter the same password again when asked to verify it.

```
C:\wamp64\bin\apache\apache2.4.65\bin>openssl genrsa -des3 -out server.key 2048
Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
```

The password will not appear while typing. This is normal.

Step 8 — Create an unencrypted private key for Apache

openssl rsa -in server.key -out server.pem

This creates a private key that Apache can load without asking for a password during every restart.

Enter the same password used in the previous step: Admin123

```
C:\wamp64\bin\apache\apache2.4.65\bin>openssl rsa -in server.key -out server.pem
Enter pass phrase for server.key:
writing RSA key
```

A successful result should include :writing RSA key

Step 9 — Generate the Certificate Signing Request

openssl req -new -key server.pem -out server.csr -config "%OPENSSL_CONF%"

This creates the Certificate Signing Request using your private key.

Enter information similar to the following:

Country Name: PK

State or Province Name: Punjab

Locality Name: Lahore

Organization Name: CND Demo

Organizational Unit Name: Development

Common Name: localhost

Email Address: your-email@example.com

The most important field is:

Common Name: localhost

For these two questions, press **Enter** without typing anything:

A challenge password:

An optional company name:

```
C:\wamp64\bin\apache\apache2.4.65\bin>openssl req -new -key server.pem -out server.csr -config "%OPENSSL_CONF%"
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:PK
State or Province Name (full name) [Some-State]:PUNJAB
Locality Name (eg, city) []:LAHORE
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Global Tech
Organizational Unit Name (eg, section) []:Cyber
Common Name (e.g. server FQDN or YOUR name) []:localhost
Email Address []:abcxyz@example.com

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:Admin123
An optional company name []:Admin123
```

Step 10 — Create the Subject Alternative Name file

echo subjectAltName=DNS:localhost,IP:127.0.0.1>localhost.ext

```
C:\wamp64\bin\apache\apache2.4.65\bin>echo subjectAltName=DNS:localhost,IP:127.0.0.1>localhost.ext
```

This allows the certificate to work with both localhost and 127.0.0.1.

Step 11 — Check the Subject Alternative Name file

type localhost.ext

It should display:

```
C:\wamp64\bin\apache\apache2.4.65\bin>type localhost.ext
subjectAltName=DNS:localhost,IP:127.0.0.1
```

subjectAltName=DNS:localhost,IP:127.0.0.1

Step 12 — Generate the self-signed SSL certificate

openssl x509 -req -days 365 -sha256 -in server.csr -signkey server.pem -out server.crt -extfile localhost.ext

This creates a self-signed SSL certificate valid for 365 days.

A successful result should include:

```
C:\wamp64\bin\apache\apache2.4.65\bin>openssl x509 -req -days 365 -sha256 -in server.csr -signkey server.pem -out server.crt
-extfile localhost.ext
Certificate request self-signature ok
subject=C=PK, ST=PUNJAB, L=LAHORE, O=Global Tech, OU=Cyber, CN=localhost, emailAddress=abcxyz@example.com
```

Certificate request self-signature ok

Step 13 — Verify all generated files

dir server.*

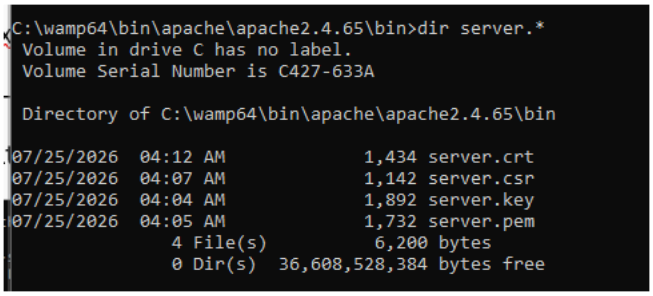
You should see:

server.key

server.pem

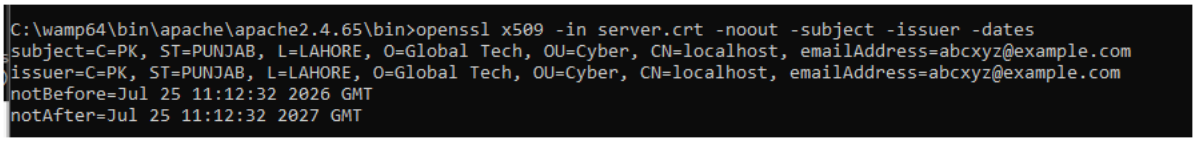
server.csr

server.crt



Step 14 — Check the certificate details

openssl x509 -in server.crt -noout -subject -issuer -dates

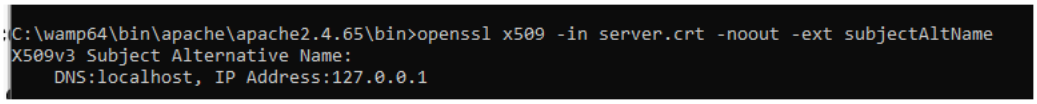


This displays the certificate subject, issuer, start date, and expiry date.

Step 15 — Verify localhost in the certificate

openssl x509 -in server.crt -noout -ext subjectAltName

The expected result is:



DNS:localhost, IP Address:127.0.0.1

Step 16 — Create the Apache SSL folder

mkdir C:\wamp64\bin\apache\apache2.4.65\conf\ssl

This creates a dedicated folder for the SSL certificate files.

```
C:\wamp64\bin\apache\apache2.4.65\bin>mkdir C:\wamp64\bin\apache\apache2.4.65\conf\ssl
A subdirectory or file C:\wamp64\bin\apache\apache2.4.65\conf\ssl already exists.
```

If Windows says the folder already exists, continue normally.

Step 17 — Copy the certificate files

copy /Y server.key C:\wamp64\bin\apache\apache2.4.65\conf\ssl\

This copies the encrypted backup key.

```
C:\wamp64\bin\apache\apache2.4.65\bin>copy /Y server.key C:\wamp64\bin\apache\apache2.4.65\conf\ssl\
        1 file(s) copied.
```

Step 18 — Confirm that the files were copied

dir C:\wamp64\bin\apache\apache2.4.65\conf\ssl

You should see:

server.key

server.pem

server.csr

server.crt

```
C:\wamp64\bin\apache\apache2.4.65\bin>dir C:\wamp64\bin\apache\apache2.4.65\conf\ssl
 Volume in drive C has no label.
 Volume Serial Number is C427-633A

 Directory of C:\wamp64\bin\apache\apache2.4.65\conf\ssl

07/25/2026  04:15 AM    <DIR>          .
07/25/2026  04:15 AM    <DIR>          ..
07/25/2026  04:04 AM             1,892 server.key
               1 File(s)          1,892 bytes
               2 Dir(s)  36,607,258,624 bytes free
```

Step 19 — Fix missing certificate files (troubleshooting)

When Step 18 was run, only server.key appeared in the ssl folder — server.pem, server.csr, and server.crt were missing.

Cause: The copy commands in Step 17 did not fully execute for all four files in the first attempt.

Fix: Re-ran the four copy commands individually from the bin folder:

copy /Y server.key C:\wamp64\bin\apache\apache2.4.65\conf\ssl\

copy /Y server.pem C:\wamp64\bin\apache\apache2.4.65\conf\ssl\

copy /Y server.csr C:\wamp64\bin\apache\apache2.4.65\conf\ssl\

copy /Y server.crt C:\wamp64\bin\apache\apache2.4.65\conf\ssl\

```
C:\wamp64\bin\apache\apache2.4.65\bin>copy /Y server.key C:\wamp64\bin\apache\apache2.4.65\conf\ssl\
        1 file(s) copied.

C:\wamp64\bin\apache\apache2.4.65\bin>copy /Y server.pem C:\wamp64\bin\apache\apache2.4.65\conf\ssl\
        1 file(s) copied.

C:\wamp64\bin\apache\apache2.4.65\bin>copy /Y server.csr C:\wamp64\bin\apache\apache2.4.65\conf\ssl\
        1 file(s) copied.

C:\wamp64\bin\apache\apache2.4.65\bin>copy /Y server.crt C:\wamp64\bin\apache\apache2.4.65\conf\ssl\
        1 file(s) copied.
```

Then re-verified with:

dir C:\wamp64\bin\apache\apache2.4.65\conf\ssl

```
C:\wamp64\bin\apache\apache2.4.65\bin>dir C:\wamp64\bin\apache\apache2.4.65\conf\ssl
 Volume in drive C has no label.
 Volume Serial Number is C427-633A

 Directory of C:\wamp64\bin\apache\apache2.4.65\conf\ssl

07/25/2026  05:10 AM    <DIR>          .
07/25/2026  05:10 AM    <DIR>          ..
07/25/2026  04:12 AM             1,434 server.crt
07/25/2026  04:07 AM             1,142 server.csr
07/25/2026  04:04 AM             1,892 server.key
07/25/2026  04:05 AM             1,732 server.pem
               4 File(s)          6,200 bytes
               2 Dir(s)  36,080,574,464 bytes free
```

Result: All 4 files (server.key, server.pem, server.csr, server.crt) confirmed present before proceeding to enable Apache modules.

Complete Command List

Run this complete sequence one command at a time:

```
cd /d C:\wamp64\bin\apache\apache2.4.65\bin

set "OPENSSL_CONF=C:\wamp64\bin\php\php8.3.28\extras\ssl\openssl.cnf"

echo %OPENSSL_CONF%

if exist "%OPENSSL_CONF%" (echo CONFIG FILE FOUND) else (echo CONFIG FILE NOT FOUND)

openssl version

del /Q server.key server.pem server.csr server.crt localhost.ext 2>nul

openssl genrsa -des3 -out server.key 2048

openssl rsa -in server.key -out server.pem

openssl req -new -key server.pem -out server.csr -config "%OPENSSL_CONF%"

echo subjectAltName=DNS:localhost,IP:127.0.0.1>localhost.ext

type localhost.ext

openssl x509 -req -days 365 -sha256 -in server.csr -signkey server.pem -out server.crt -extfile
localhost.ext

dir server.*

openssl x509 -in server.crt -noout -subject -issuer -dates

openssl x509 -in server.crt -noout -ext subjectAltName

mkdir C:\wamp64\bin\apache\apache2.4.65\conf\ssl

copy /Y server.key C:\wamp64\bin\apache\apache2.4.65\conf\ssl\

copy /Y server.pem C:\wamp64\bin\apache\apache2.4.65\conf\ssl\

copy /Y server.csr C:\wamp64\bin\apache\apache2.4.65\conf\ssl\

copy /Y server.crt C:\wamp64\bin\apache\apache2.4.65\conf\ssl\

dir C:\wamp64\bin\apache\apache2.4.65\conf\ssl
```

After Completing the Commands

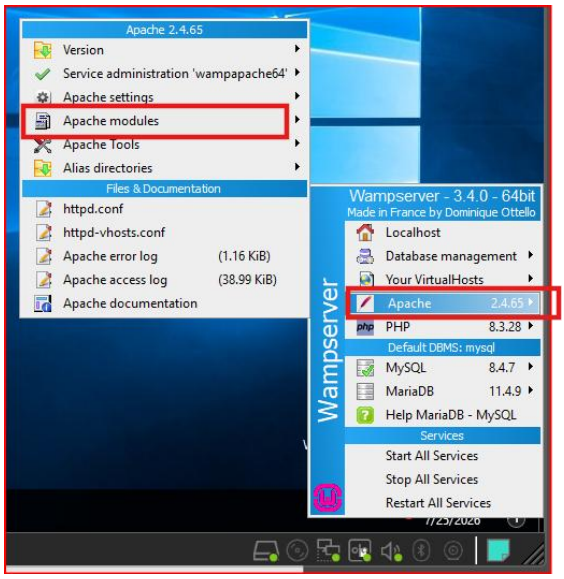
Restart your VM & start your WAMP Server



1. Enable the required Apache modules

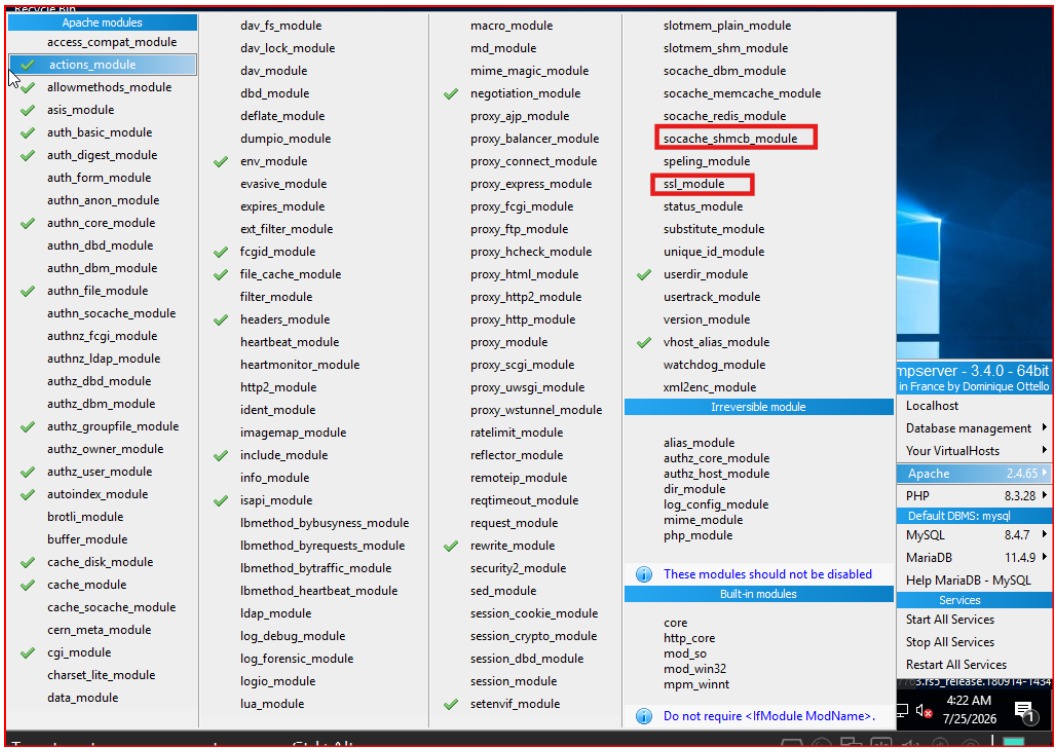
Click the WampServer tray icon and enable:

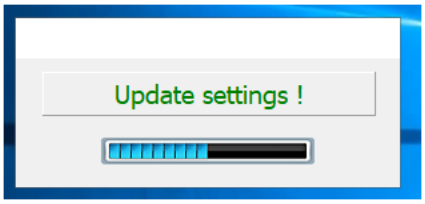
Apache → Apache modules → ssl_module

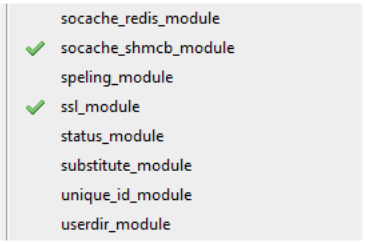


Also enable:

Apache → Apache modules → socache_shmcb_module





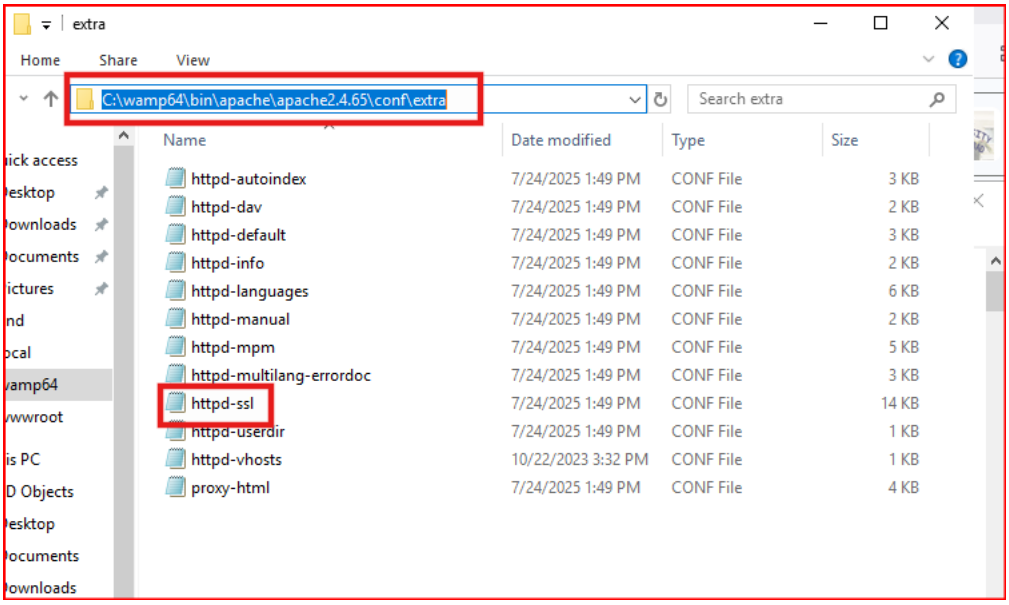


Wait for the WampServer icon to return to green.

2. Open the SSL configuration file

Open:

C:\wamp64\bin\apache\apache2.4.65\conf\extra\httpd-ssl.conf



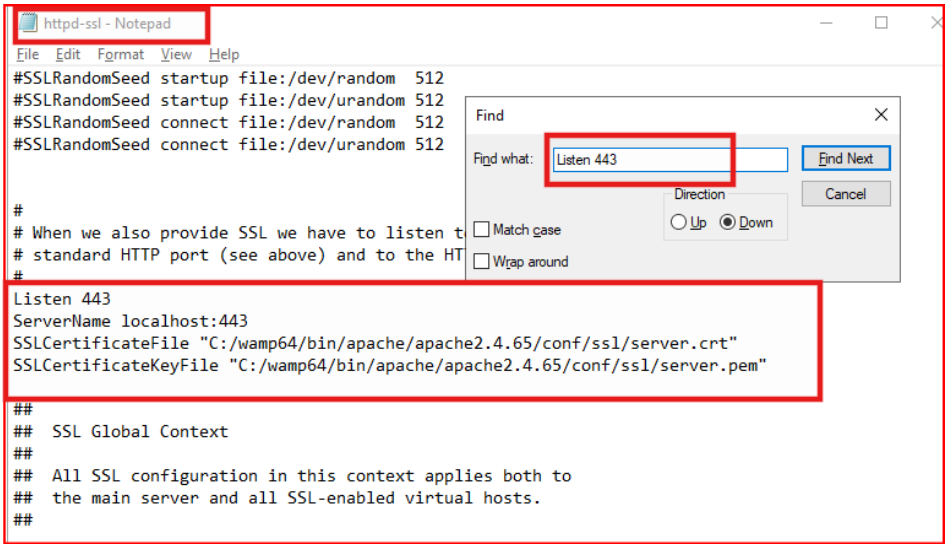
Set these lines:

```
Listen 443

ServerName localhost:443

SSLCertificateFile "C:/wamp64/bin/apache/apache2.4.65/conf/ssl/server.crt"

SSLCertificateKeyFile "C:/wamp64/bin/apache/apache2.4.65/conf/ssl/server.pem"
```



Save the file.

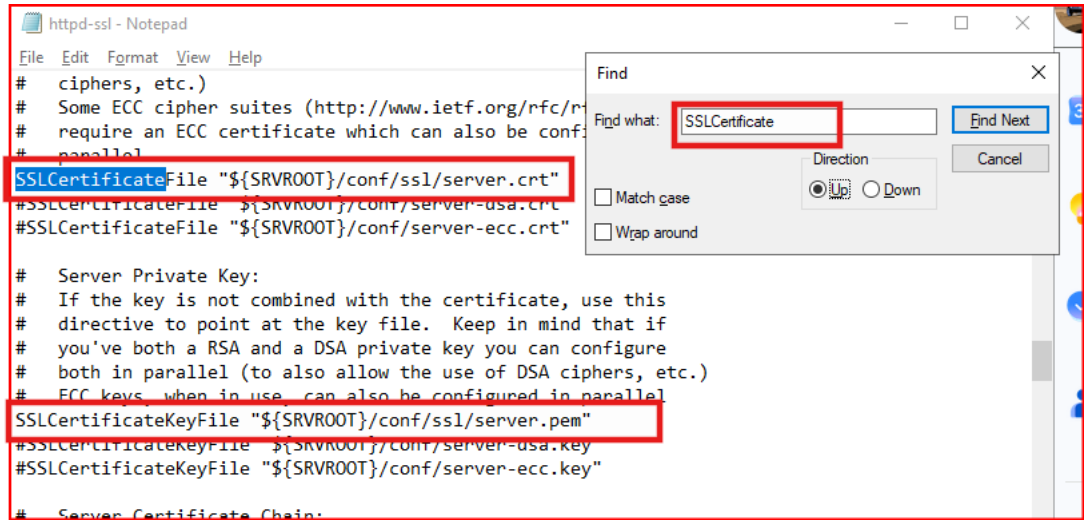
Now search for SSL Certificate

1. Change the first line to:

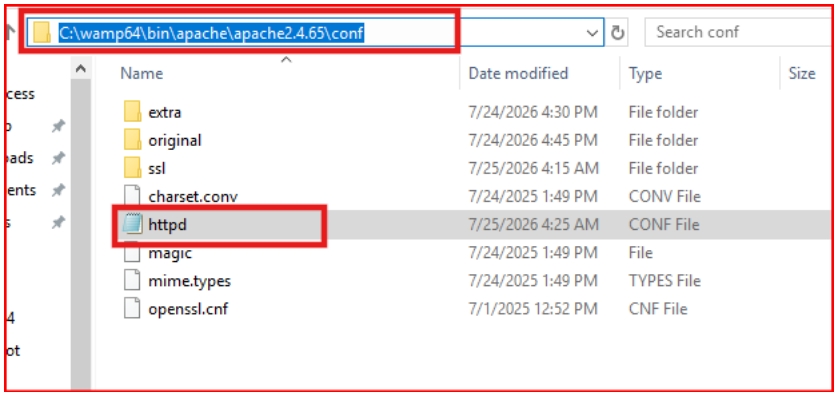
```
SSL CertificateFile "${SRVROOT}/conf/ssl/server.crt"
```

2. Change the second line to:

```
SSLCertificateKeyFile "${SRVROOT}/conf/ssl/server.pem"
```



1. Open **C:\wamp64\bin\apache\apache2.4.65\conf\httpd.conf**



2. Press **Ctrl+F**, search for: ssl
3. Look specifically for a line like:

Include conf/extra/httpd-ssl.conf

Now remove the **#** so that it wouldn't be a comment anymore (Remove #)

```
#Secure (SSL/TLS) connections
Include conf/extra/httpd-ssl.conf

# Note: The following must must be present to support
#       starting without SSL on platforms with no /dev/random equivalent
#       but a statically compiled-in mod_ssl.
#
<IfModule ssl_module>
SSLRandomSeed startup builtin
SSLRandomSeed connect builtin
</IfModule>

Include "${INSTALL_DIR}/alias/*.conf"
```

Ctrl + S and save it

Wrong DocumentRoot and ServerName

DocumentRoot "${SRVROOT}/htdocs"

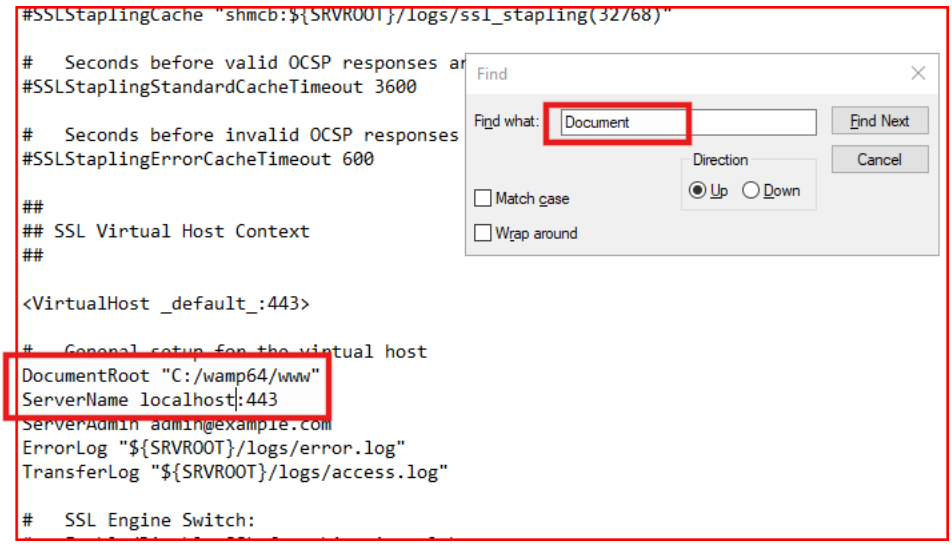
ServerName www.example.com:443

This points to a folder that doesn't even exist in your WampServer setup (htdocs isn't used by WampServer — it uses www), and the ServerName is still the placeholder.

Fix: Change both to:

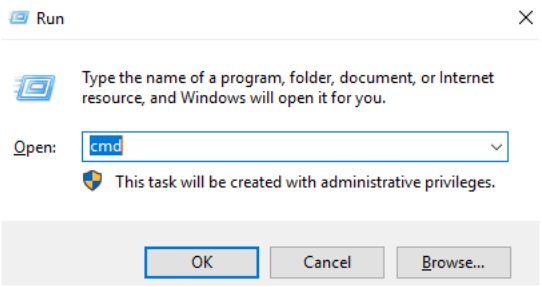
DocumentRoot "C:/wamp64/www"

ServerName localhost:443



3. Test the Apache configuration

Return to Administrator Command Prompt and run:

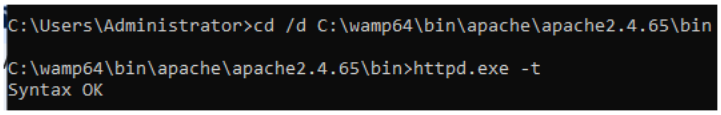


cd /d C:\wamp64\bin\apache\apache2.4.65\bin

httpd.exe -t

The correct result is:

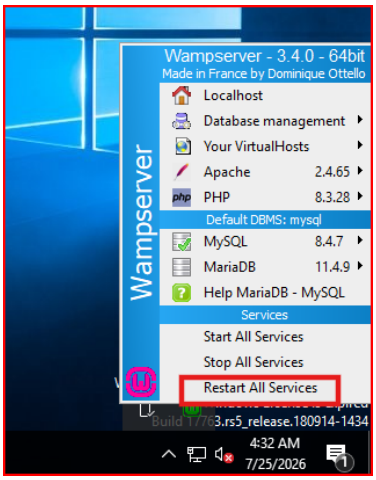
Syntax OK



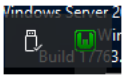
4. Restart WampServer

Use:

WampServer tray icon → Restart All Services



Wait until the icon becomes green.

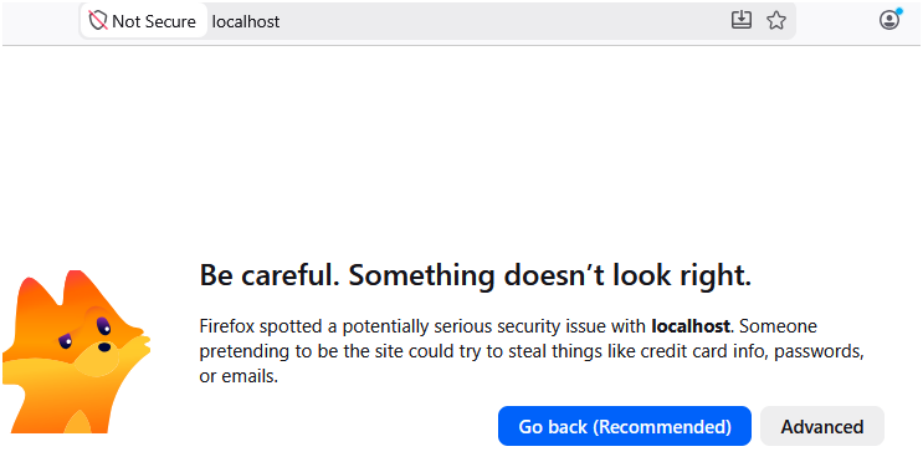


5. Test HTTPS

Open:

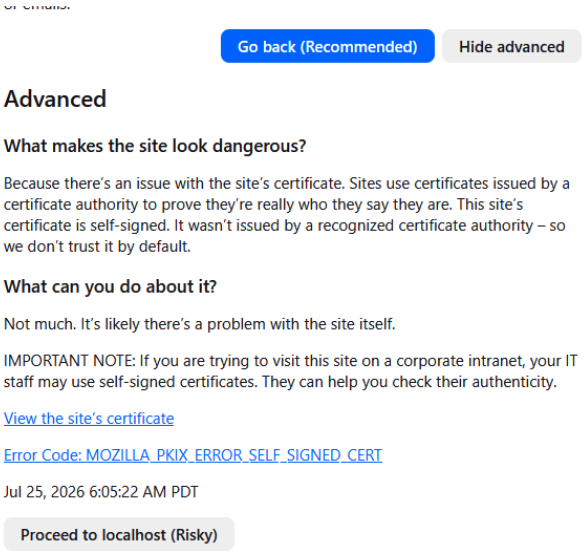
https://localhost/

A browser security warning is normal because this is a self-signed local certificate.

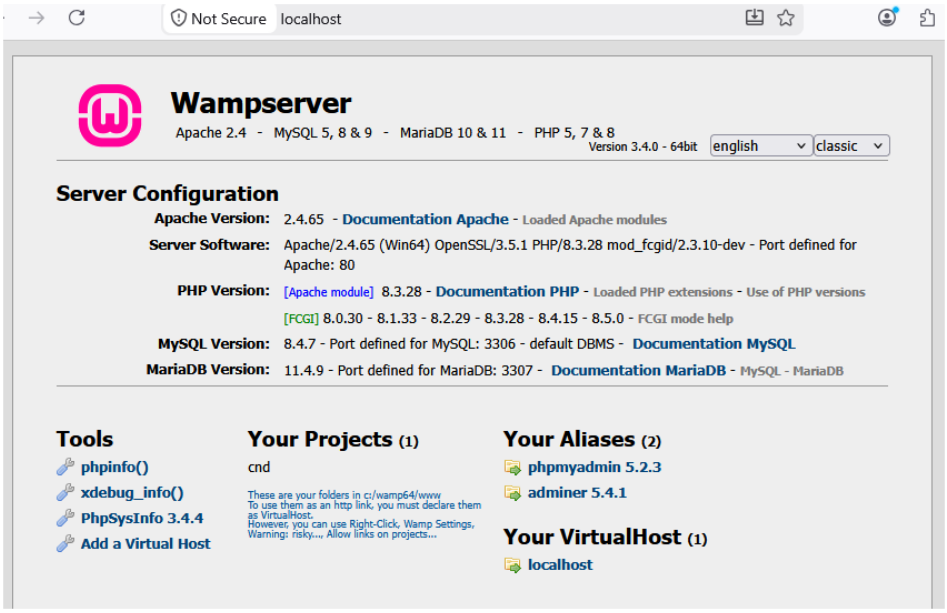


A **certificate warning page** (red screen: "Your connection is not private" / "NET::ERR_CERT_AUTHORITY_INVALID") — this is actually the **expected, correct result** since it's a self-signed certificate not trusted by the browser.

Proceed to localhost



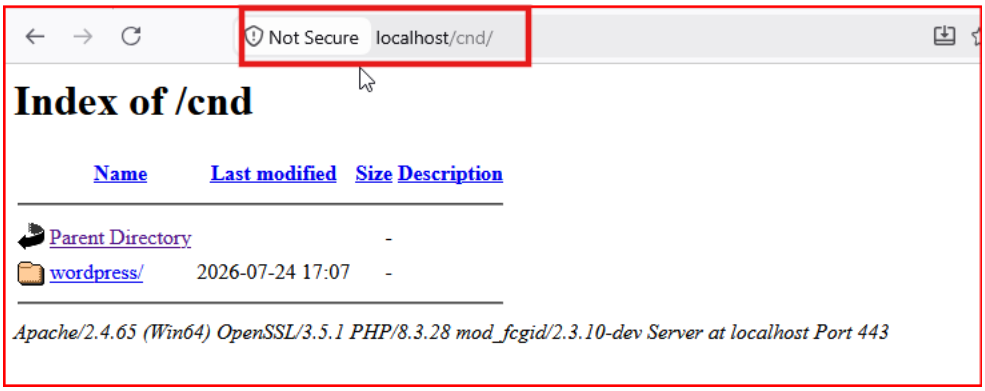
WAMP Server Running

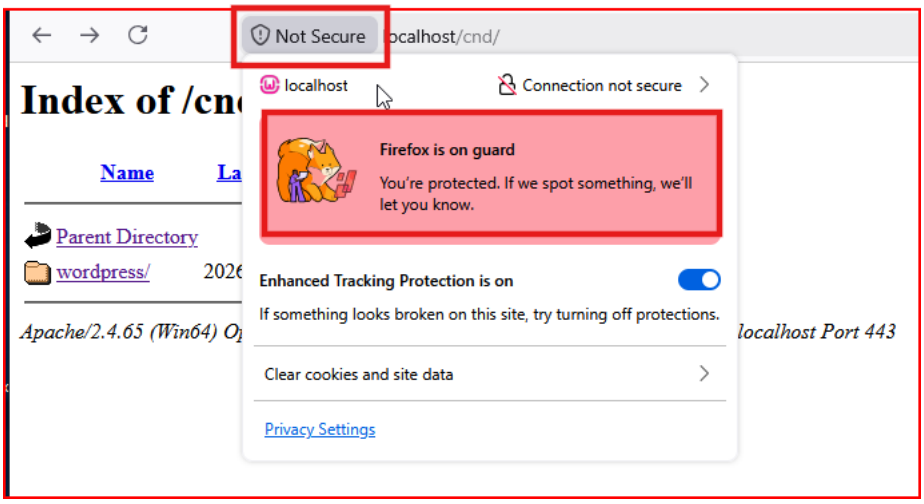


Test HTTPS explicitly

Type this directly in the address bar:

https://localhost/cnd





the address bar shows https:// with the browser's standard self-signed-certificate warning

Secure Email Communication Using PGP

Executive Summary

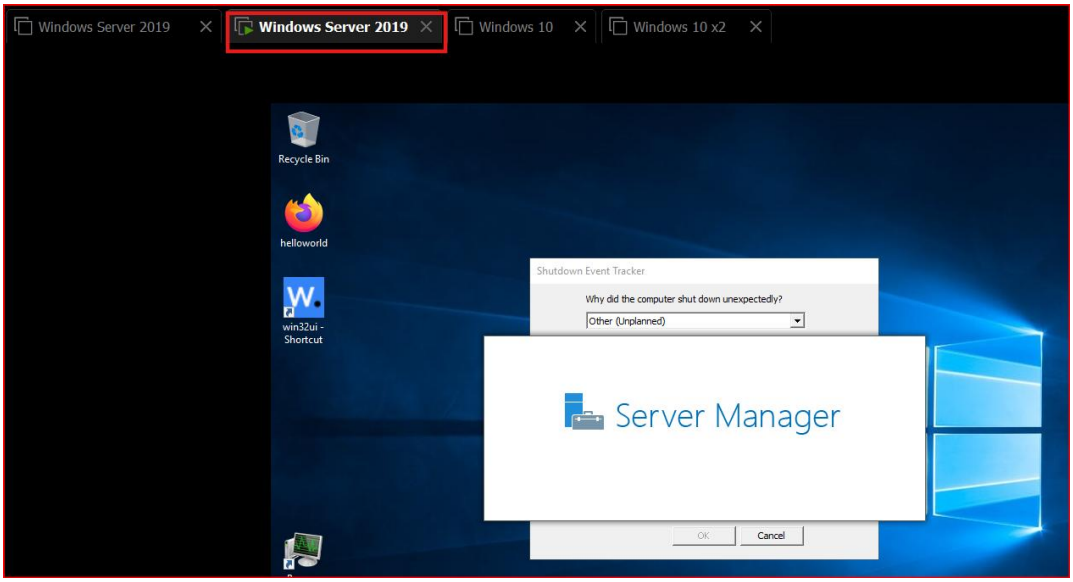
This lab demonstrates how to protect message confidentiality using OpenPGP encryption. Using Gpg4win and Kleopatra on the WebServer virtual machine, you create a public-private key pair, export the public key, back up the private key, encrypt a sample message, and successfully decrypt it. The lab confirms that only the holder of the correct private key and passphrase can recover the original content.

In this **Lab,** we will use **Gpg4win** and its certificate-management application, **Kleopatra**, to:

1. Create an **OpenPGP public-and-private key pair**.
2. Export the public key.
3. back up the private or secret key.
4. Encrypt a sample message.
5. Decrypt the encrypted message.
6. Confirm that the original text is corrected.

This lab uses only the **WebServer virtual machine**. (Server 2)

Start the Web- Server Virtual Machine



Creating Two Desktop Folders

PGP keys

Public keys



Purpose of the folders

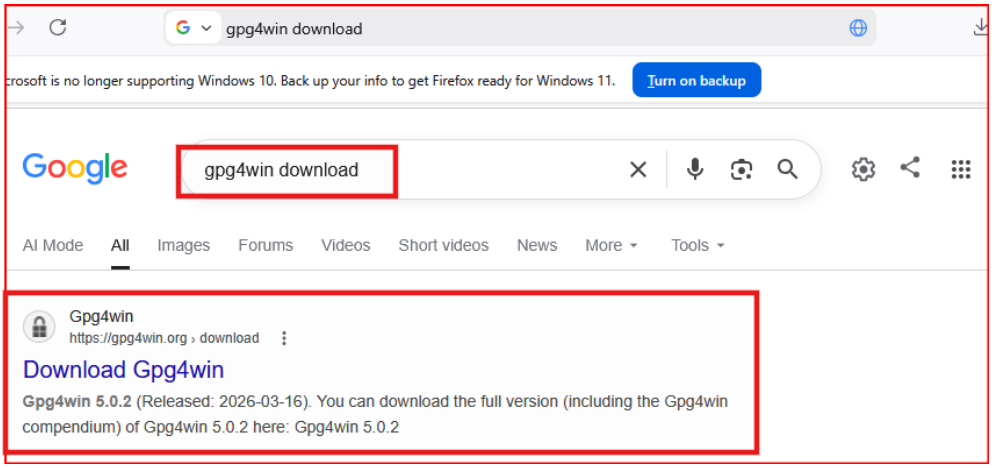
- **PGP keys** can hold the private-key backup and related cryptographic files.
- **Public keys** can hold exported public certificates that may safely be shared with another user.

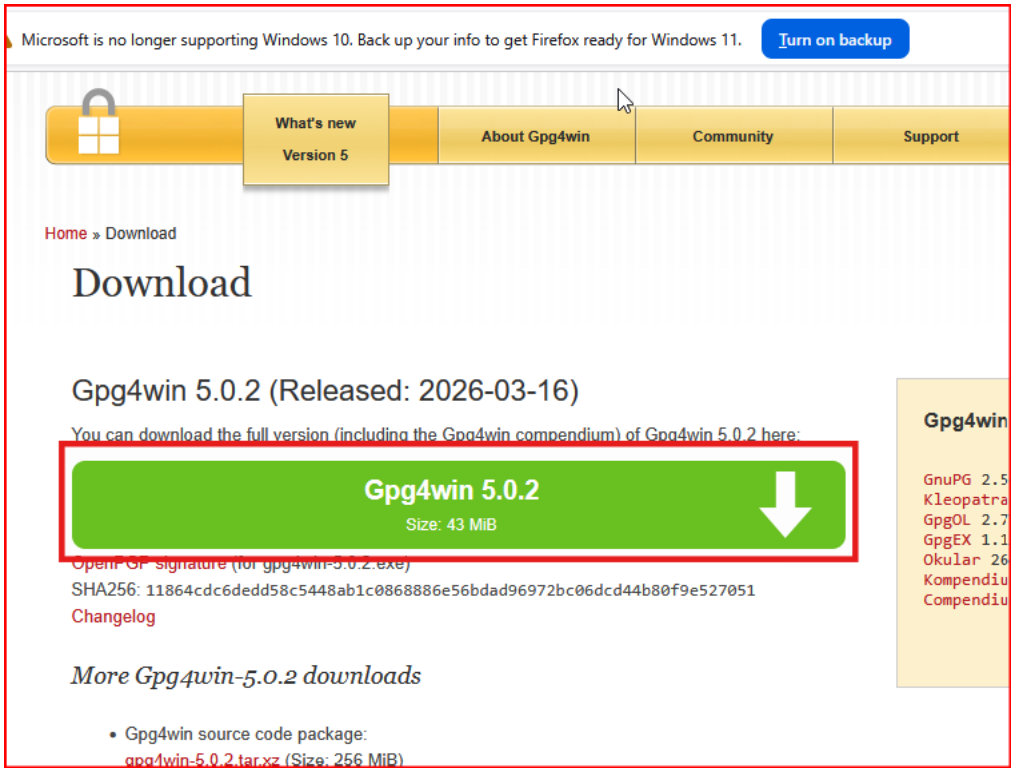
Note: Do not place normal personal documents inside the private-key folder.

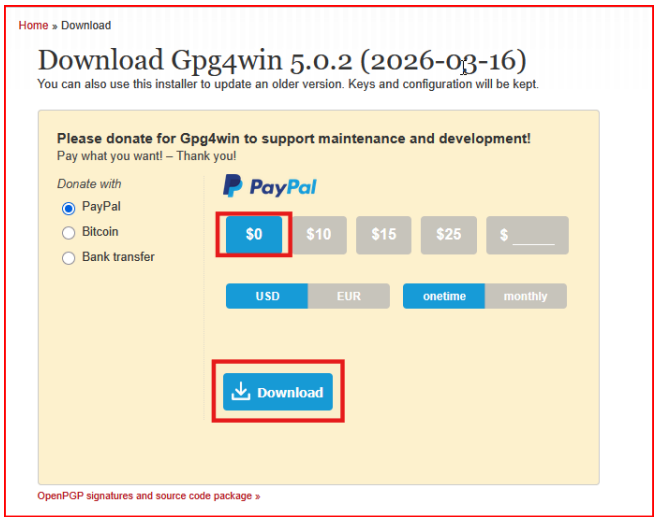
Download Gpg4win

Open only the result belonging to:

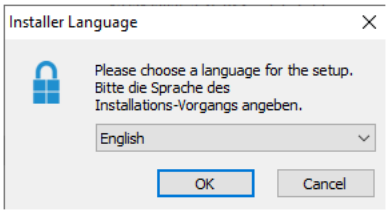
gpg4win.org







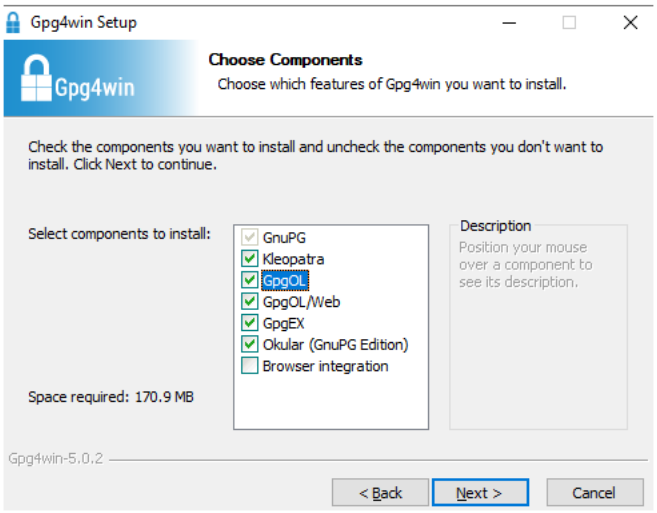
Launch Gpg4win



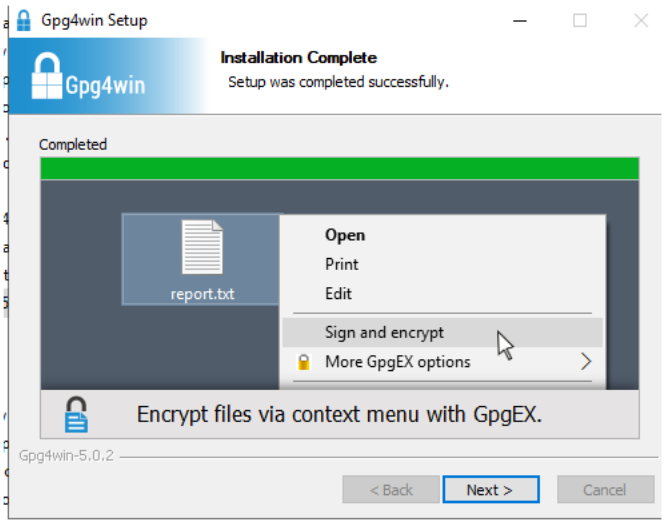


Keep these components selected (default)

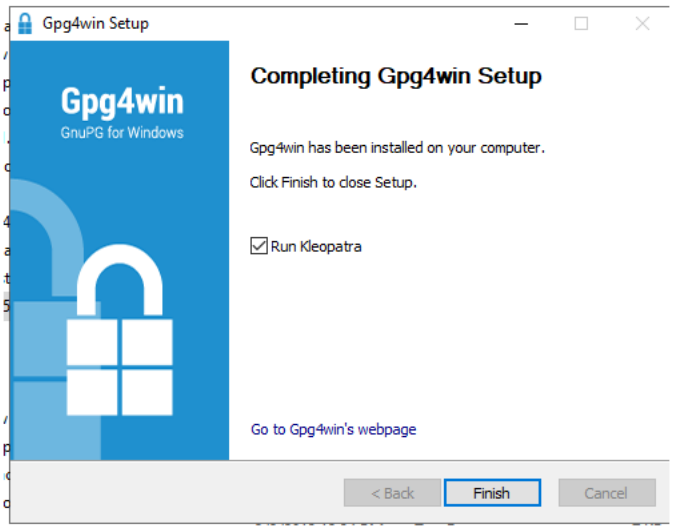
Click **Next** → **Install**.



When installation finishes, Click Next



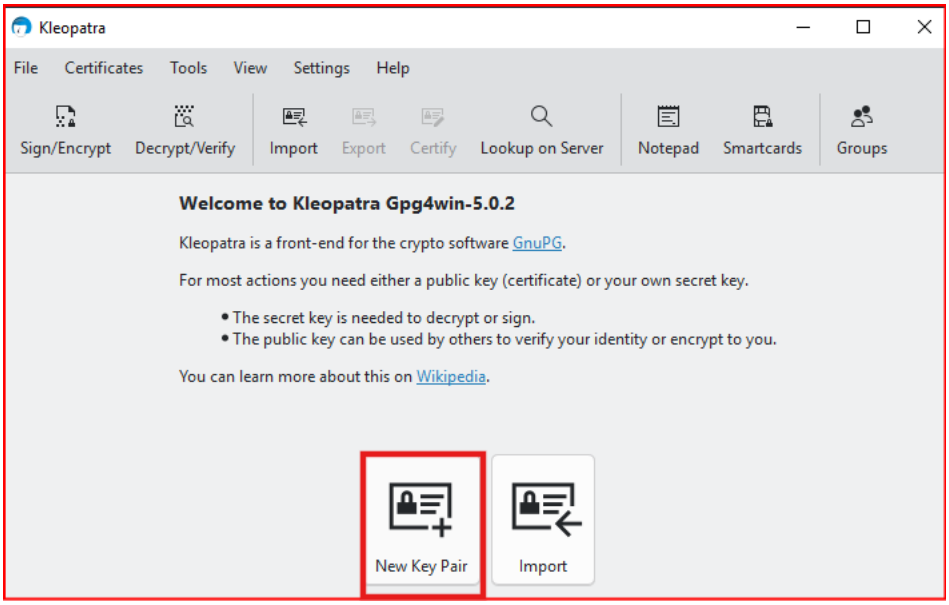
click **Finish**.



Kleopatra installed

Creating the PGP Key Pair

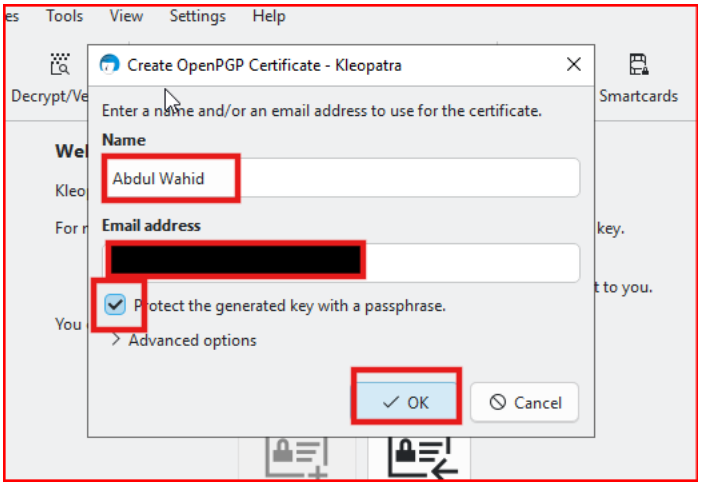
Click **New Key Pair**.



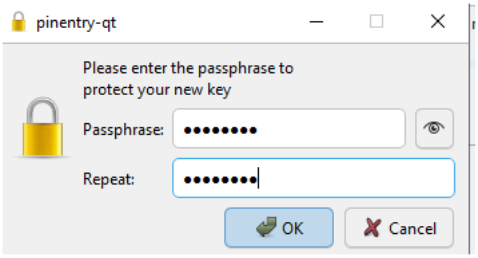
Enter name & mail

Keep **Protect the generated key with a passphrase** enabled.

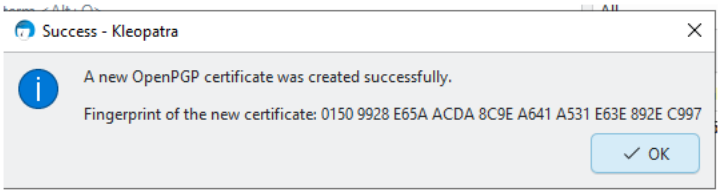
Click OK



Create a password (e.g Admin123)



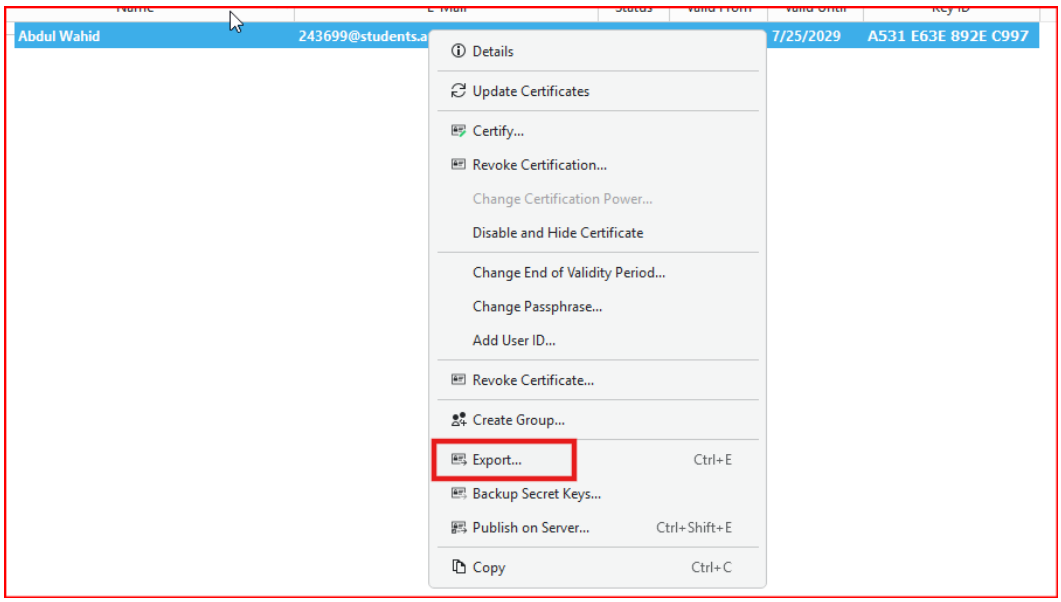
Success — your OpenPGP key pair was created correctly.



Export the Public Key

In the Kleopatra certificate list, select your newly created certificate.

- Right-click the certificate.
- Click **Export**.

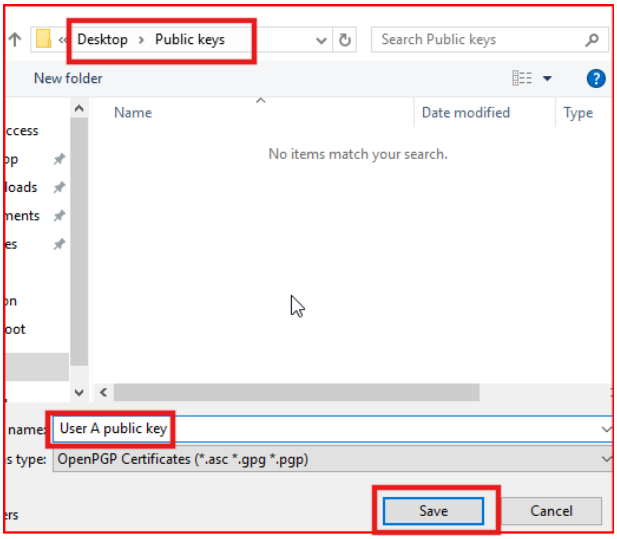


1. Select the desktop folder:

Public keys

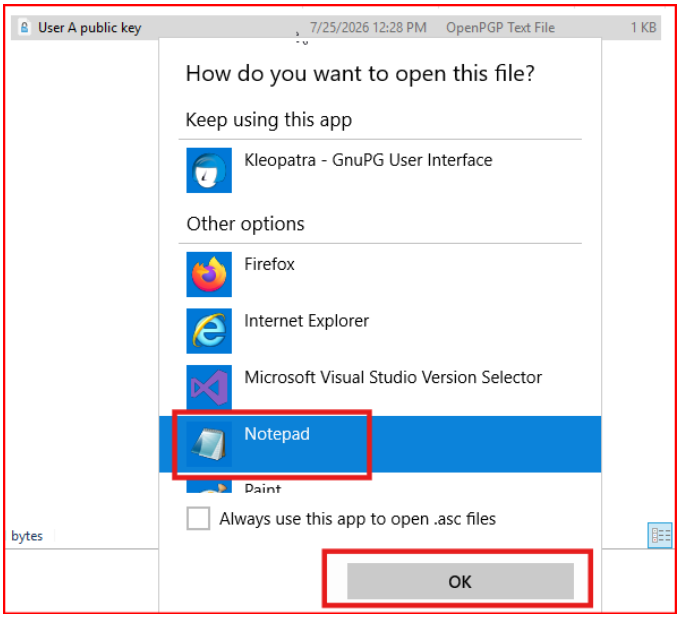
Enter a filename such as:

User A Public Key -> Click **Save**.



3. Open the **Public keys** folder from the desktop.
4. Right-click the saved public-key file and select:

Open with → Notepad



6. Confirm that the file contains encrypted-looking text beginning with:

-----BEGIN PGP PUBLIC KEY BLOCK-----

and ending with:

-----END PGP PUBLIC KEY BLOCK-----

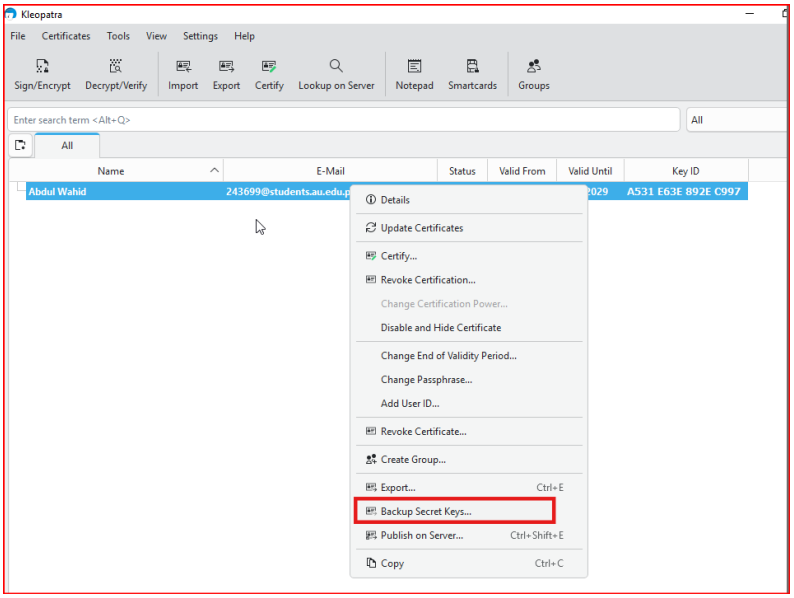
This confirms that the public key was exported correctly.



Back Up the Private Key

1. Return to **Kleopatra**.
2. Select your certificate from the certificate list.
3. Right-click the certificate and select:

Backup Secret Keys



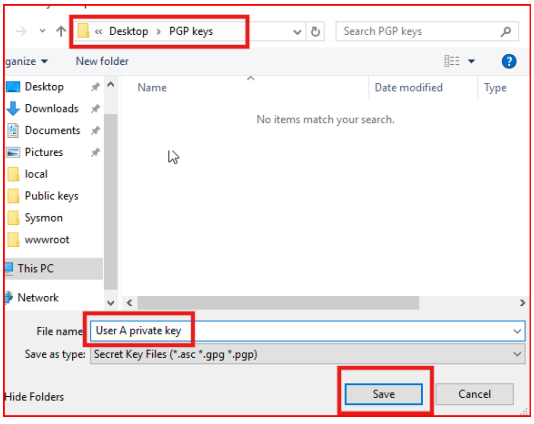
In the save window, open:

Desktop → PGP keys

Enter this filename:

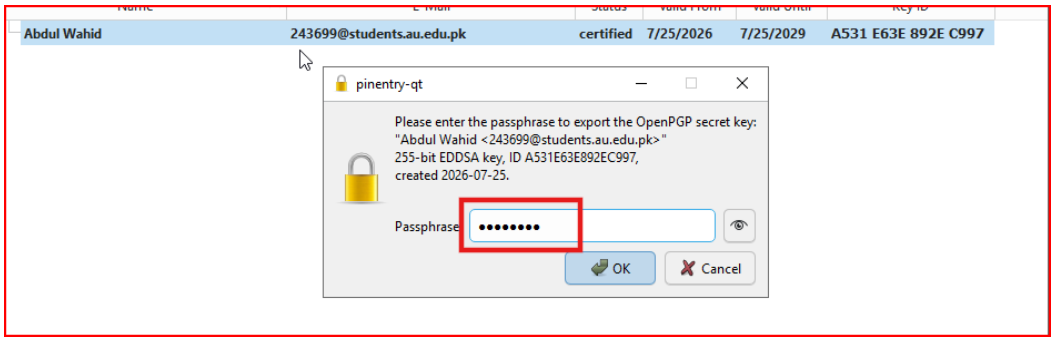
User A private key

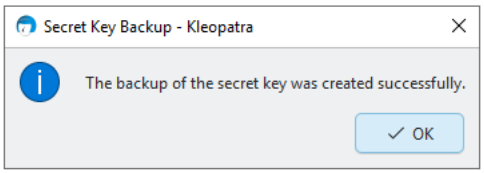
Click **Save**.



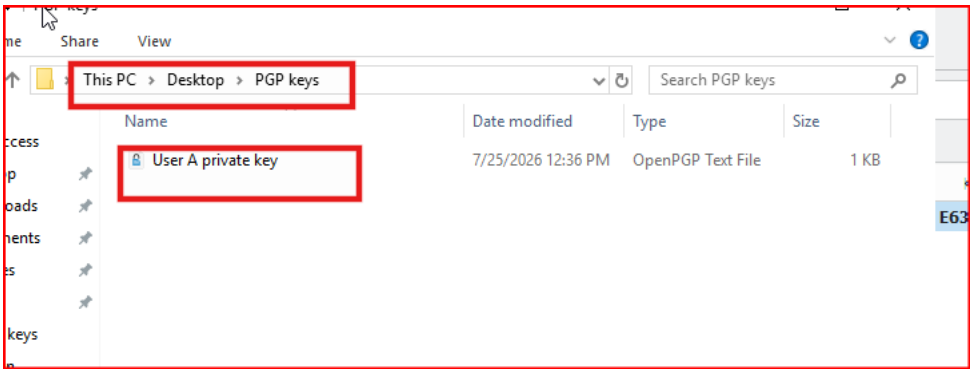
Enter password Admin123

Click OK





Open the desktop **PGP keys** folder and confirm that the private-key backup file is present.



Your public key and private-key backup are now completed. Keep the private-key file protected and do not share it. The manual performs these exports after creating the OpenPGP certificate.

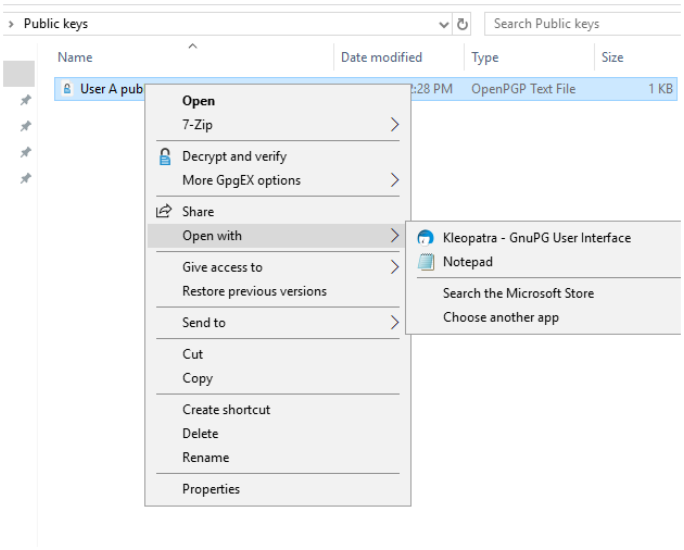
Create the Public Key Text File

1. Open:

Desktop → Public keys

2. Right-click **User A public key** and select:

Open with → Notepad



3. Confirm that the text starts with:

-----BEGIN PGP PUBLIC KEY BLOCK-----

and ends with:

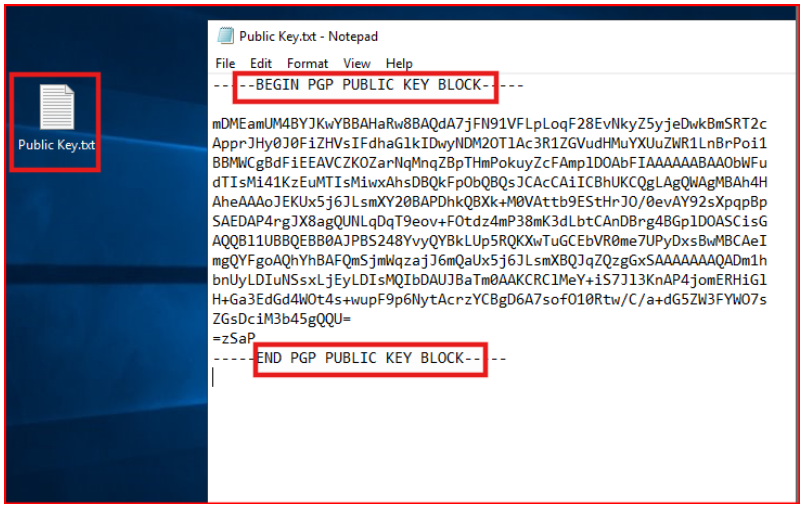
-----END PGP PUBLIC KEY BLOCK-----



Now copy all this text

8. Save it on the **Desktop** with this name:

Public Key.txt

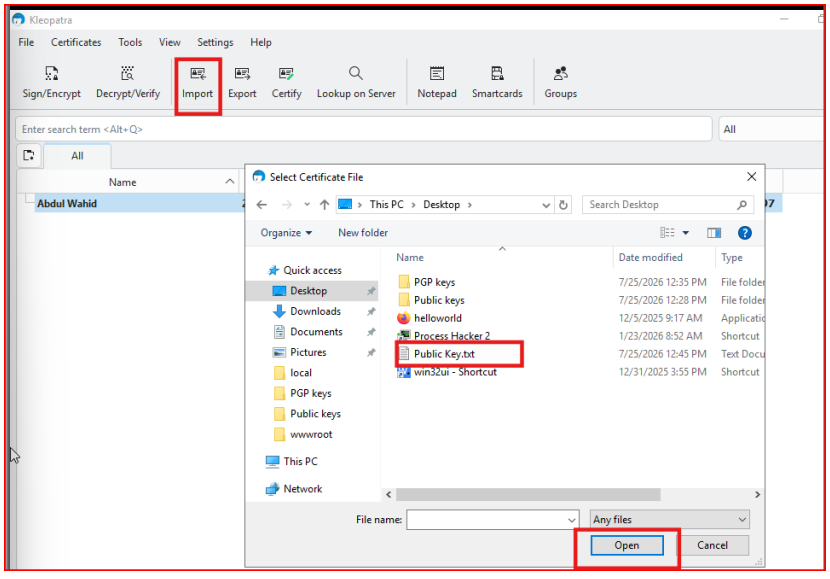


Import the Public Key

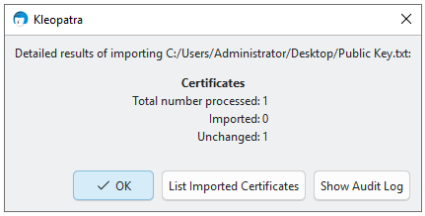
1. Return to **Kleopatra**.
2. Click **Import** from the top toolbar.
3. Select:

Desktop → Public Key.txt

- Click **Open**.



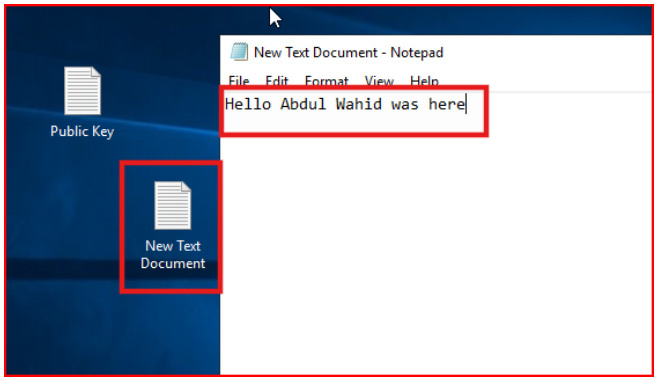
An import-results window will appear. Click **OK** or **Close**.



Encrypt the Test Message

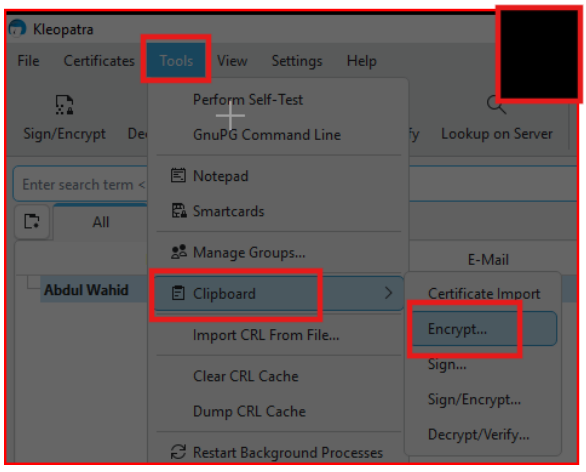
1. Open a new Notepad or Notepad++ document.
2. Type:

Hello World

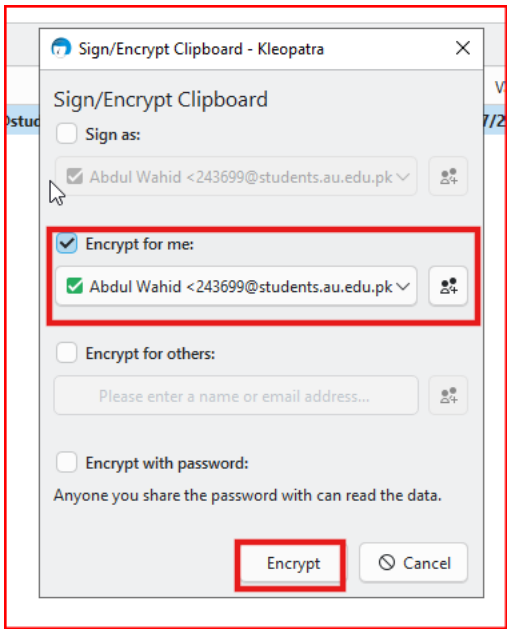


3. Select and copy it:
4. Return to **Kleopatra**.
5. Open:

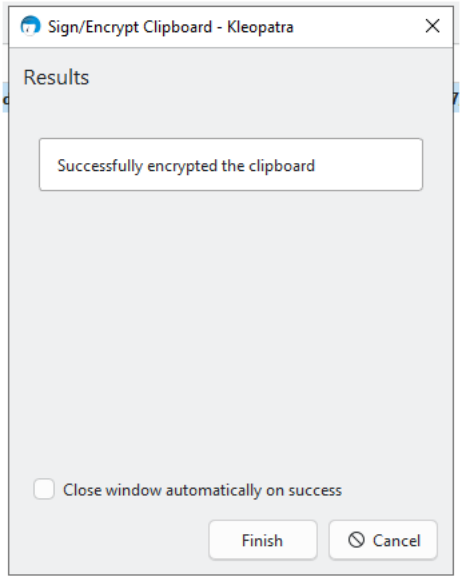
Tools → Clipboard → Encrypt



6. In the **Encrypt Mail Message** window, :
7. Your settings should be:
   ☐ Sign as
   ☑ Encrypt for me: Abdul Wahid
   ☐ Encrypt for others
   ☐ Encrypt with password
8. Now click **Encrypt**.



Click Finish



Paste the encrypted message

After Kleopatra reports that encryption succeeded:

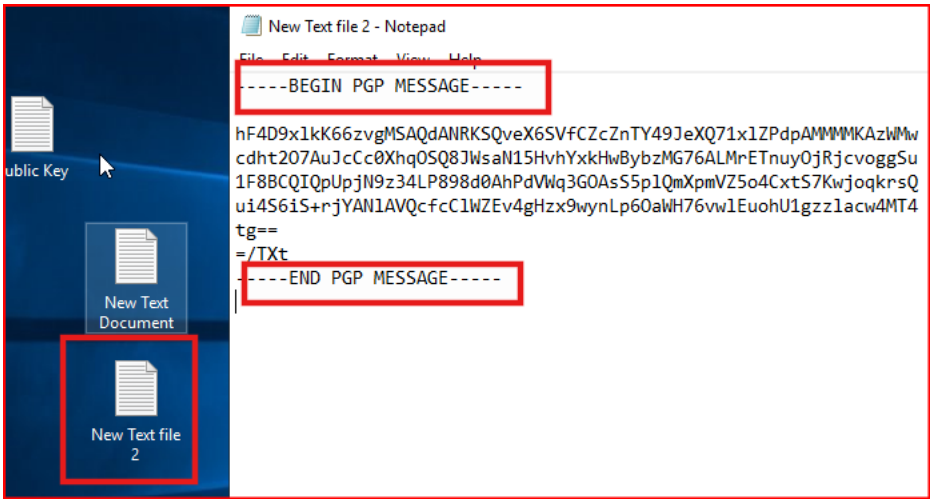
1. Click **OK** or **Finish**.
2. Open a new Notepad window.
3. Press: Ctrl + V

You should see text similar to:

-----BEGIN PGP MESSAGE-----

Encrypted characters appear here

-----END PGP MESSAGE-----



This confirms that the clipboard message was encrypted successfully.

Decrypt the Message

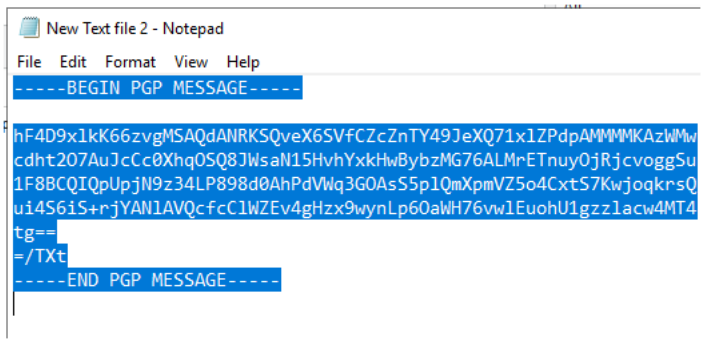
4. Copy the complete encrypted block

In Notepad:

1. Select everything from BEGIN PGP MESSAGE through END PGP MESSAGE.
2. Press:

Ctrl + A

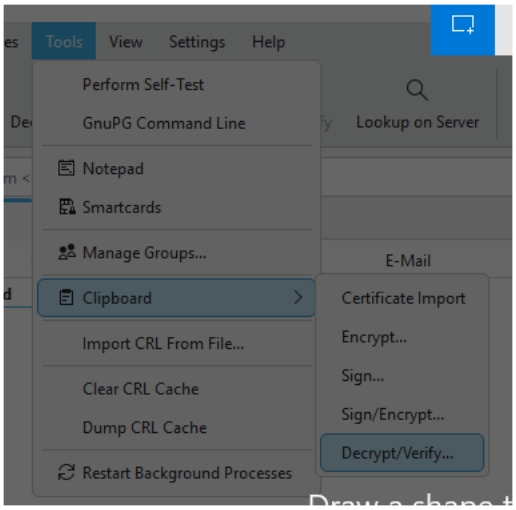
Ctrl + C



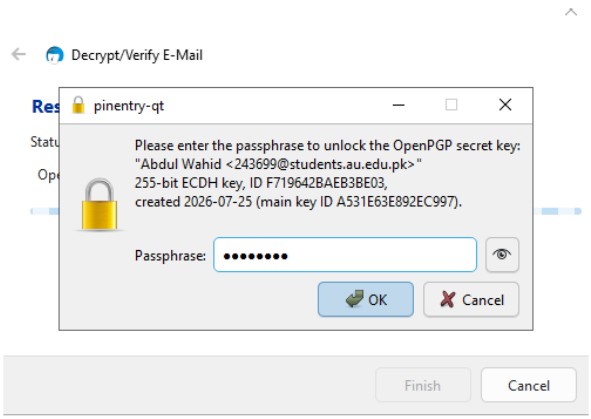
5. Decrypt through Kleopatra

Return to Kleopatra and select:

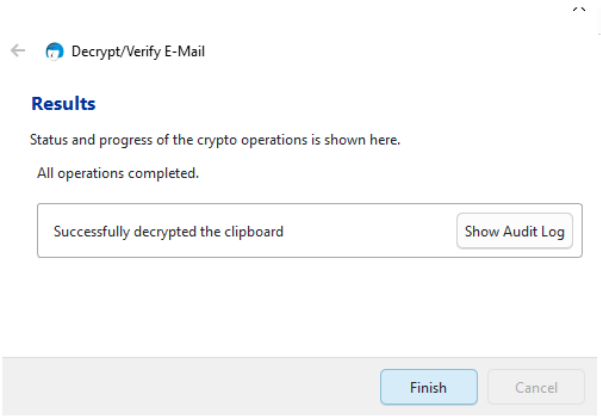
Tools → Clipboard → Decrypt/Verify



Enter your OpenPGP passphrase when requested, then click **OK**.



Wait until Kleopatra shows that decryption succeeded.

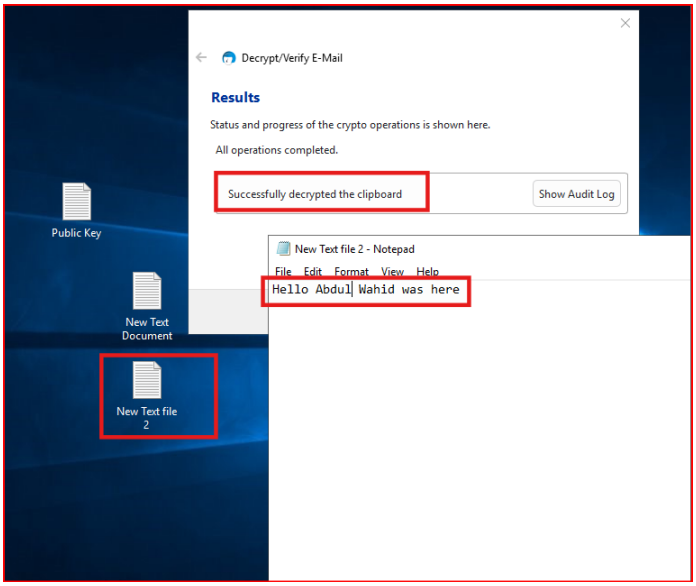


6. Check the recovered message

1. Open a new Notepad window.
2. Press:

Ctrl + V

The original text should appear:



Conclusion

- Created a public key for encryption and sharing.
- Created a protected private key for decryption.
- Saved the public key inside the Public keys folder.
- Saved the private-key backup inside the PGP keys folder.
- Encrypted a normal text message into a PGP message block.
- Decrypted the protected message and recovered the original text.

Demonstrated how OpenPGP protects message confidentiality by allowing only the holder of the correct private key and passphrase to decrypt the content.

Performing Data Backup Using AOMEI Backupper Standard

Executive Summary

This lab shows how to create and manage file backups using AOMEI Backupper Standard on AdminMachine-1. You prepare a test folder with a sample document, install the software, configure a file backup task, store the backup in a separate destination folder, and verify that the backup completes successfully. The exercise covers full backup creation and confirms data protection against accidental loss.

Lab Workflow

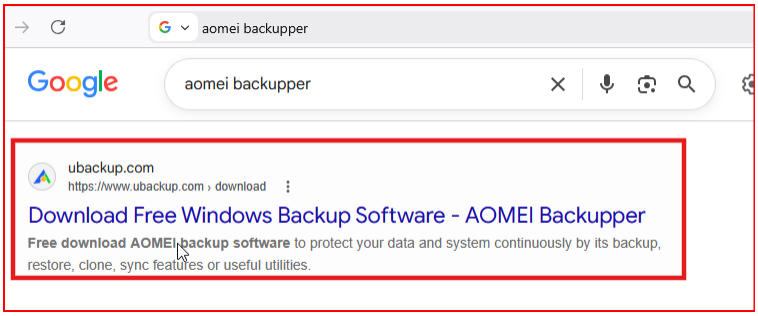
1. Prepare the AdminMachine-1 virtual machine.
2. Copy the Test Document folder to the desktop.
3. Install AOMEI Backupper Standard.
4. Create a file backup.
5. Store the backup on the F: drive.
6. configure the backup schedule.
7. Create an incremental backup.
8. Confirm that the backup files exist.
9. Uninstall AOMEI after completing the exercise.

Prepare AdminMachine-1



Open **Microsoft Edge** inside AdminMachine-1.

Search for AOMEI Backupper

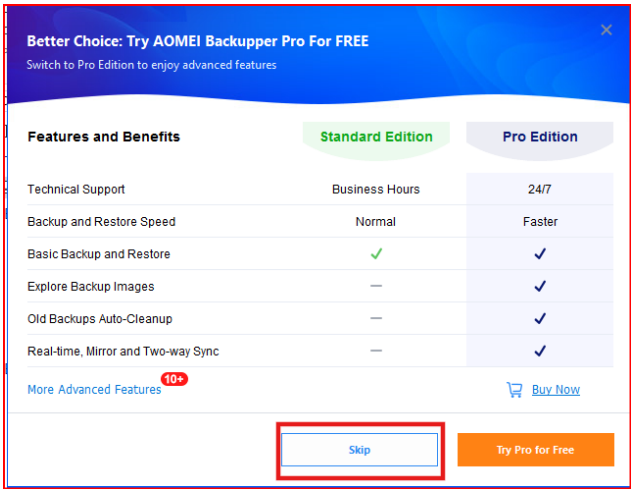


Download Freeware



LAUNCH AOMEI

Skip it



Click install



Click enjoy now



The **AOMEI Backupper** main dashboard should open successfully.

Prepare the Data and Configure the Backup

Create the Test Document Folder

1. On the **AdminMachine-1 desktop**, right-click an empty area and:

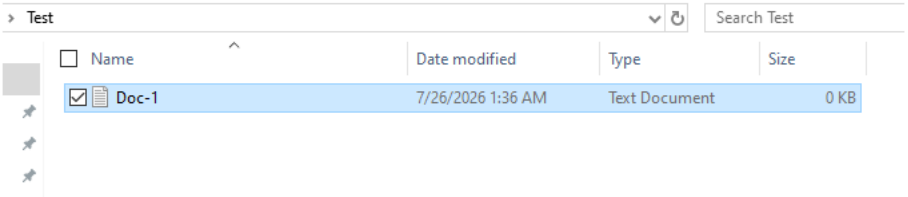
Create a New Folder name it as “Test”

Create another folder as Backup



Create a text file in the Test folder you created

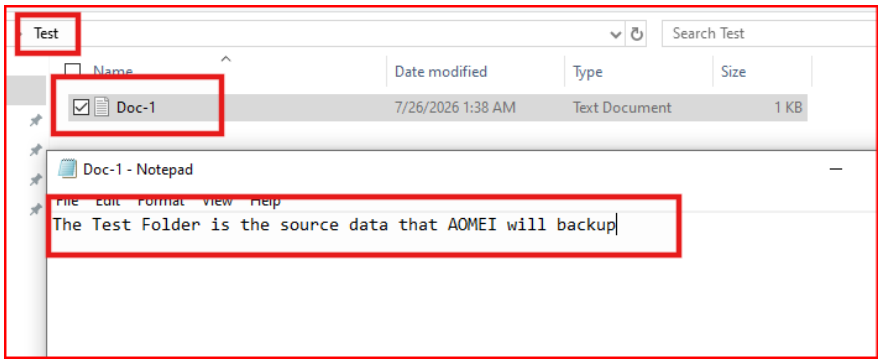
Name it as “Doc-1”



Open Doc-1.txt, type:

This is the original document for the AOMEI backup lab.

Ctrl + S

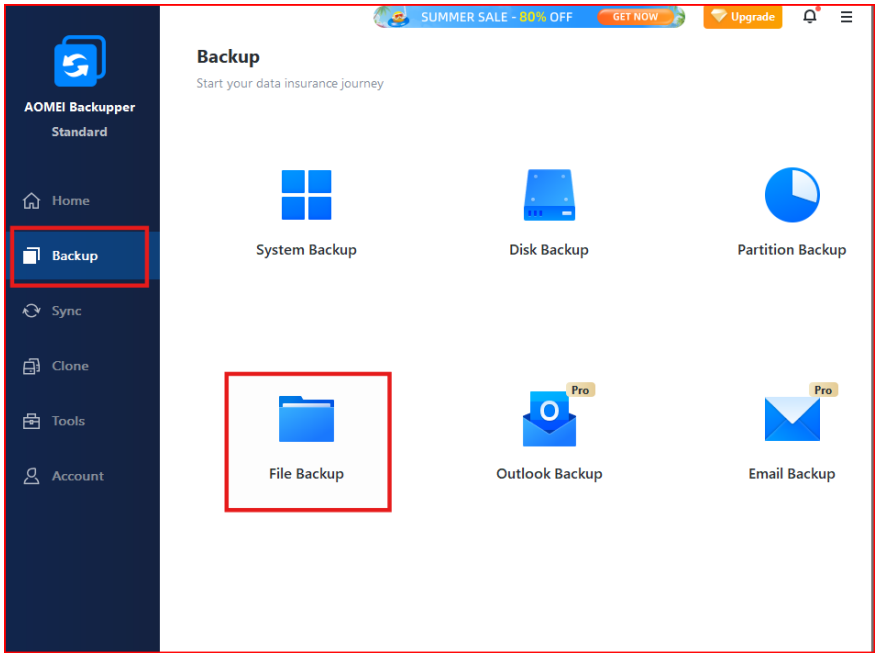


Then close Notepad.

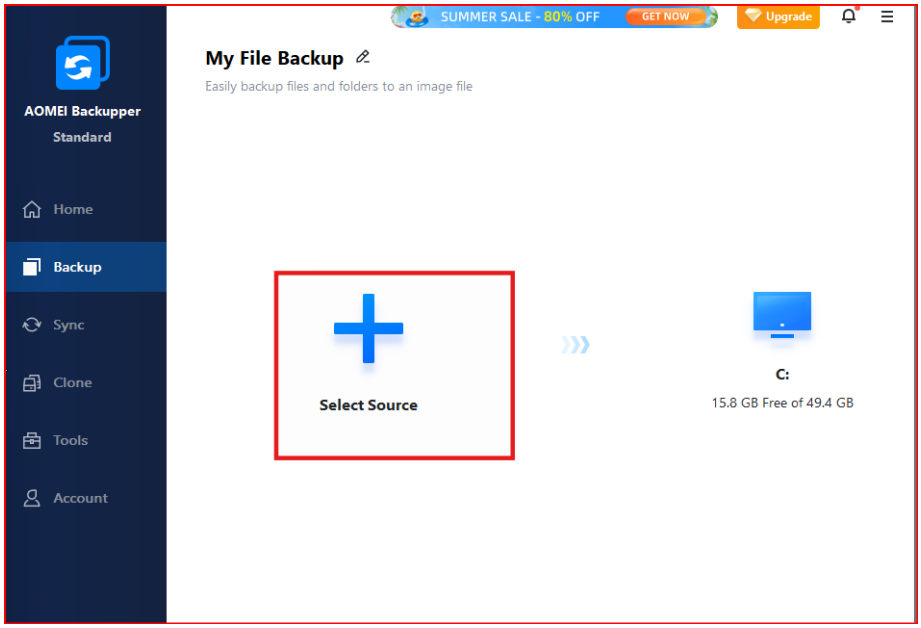
B. Configure the File Backup

1. Return to **AOMEI Backupper**.
2. Click **Backup** from the left menu.
3. Select:

File Backup



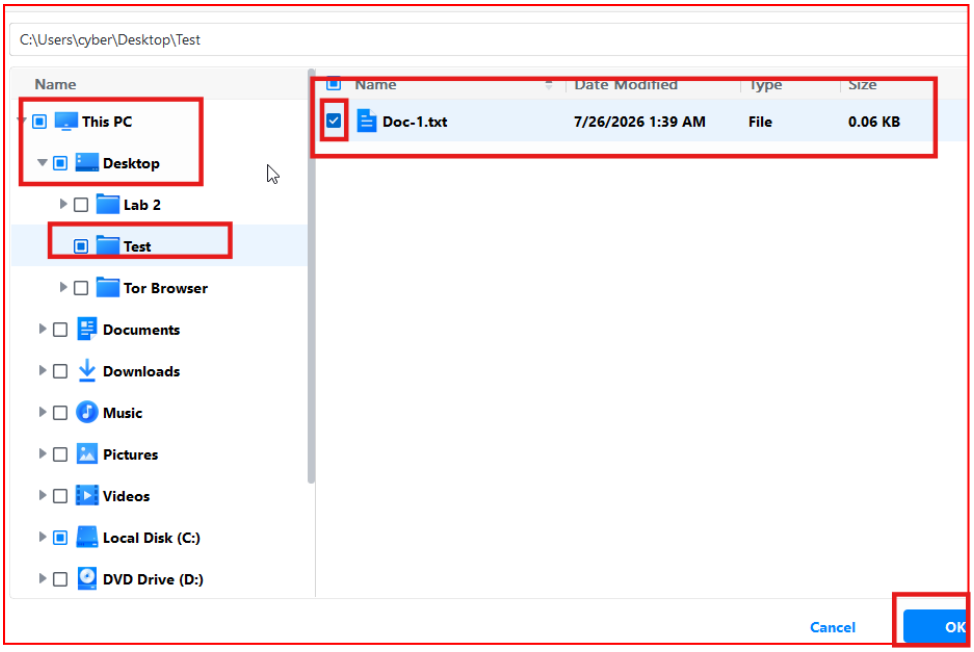
Select Source



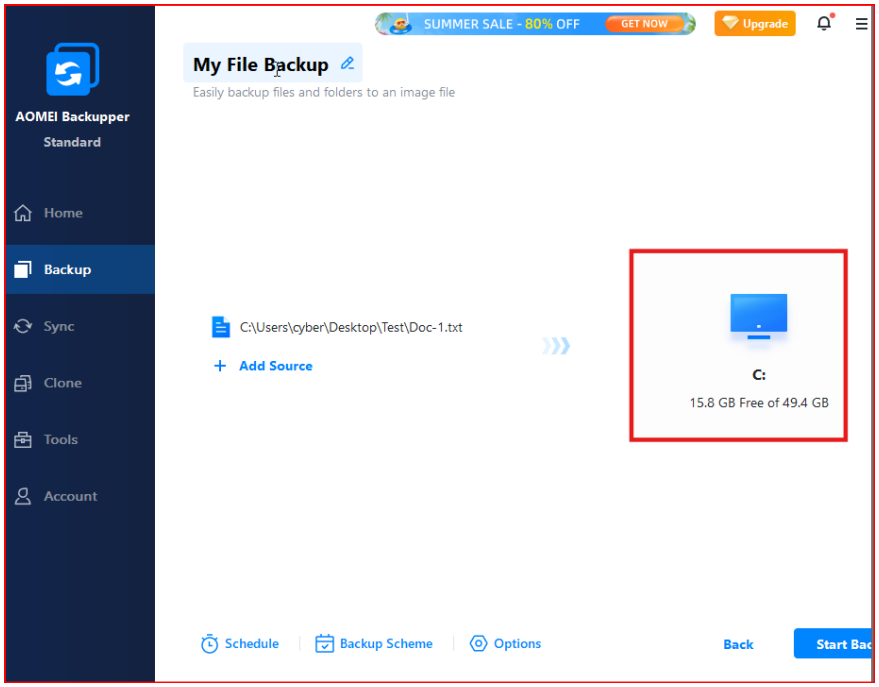
6. Browse to:

Desktop → Test Document

7. Select the folder and click **OK**.

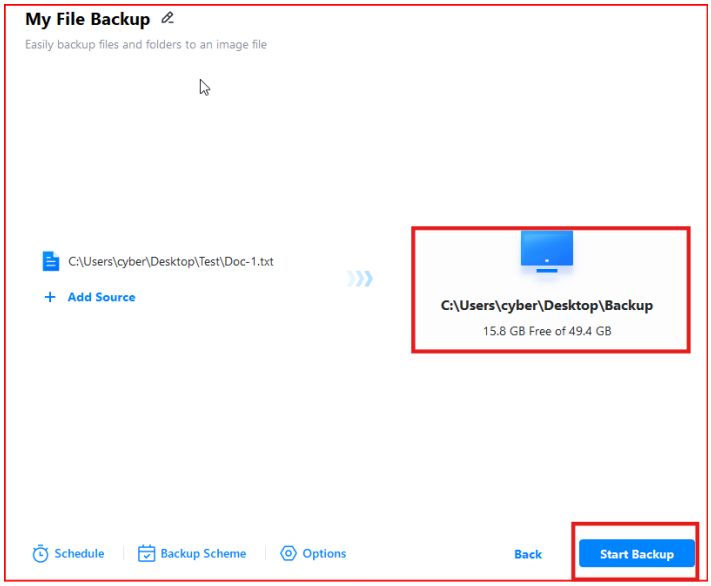


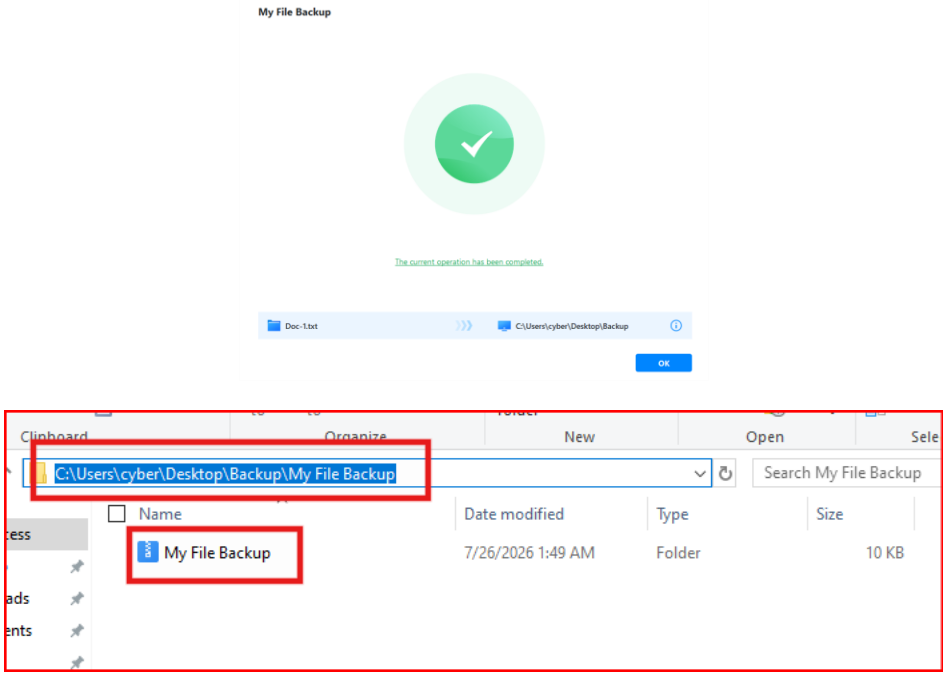
8. Click the destination-selection area on right of the source folder.



9. Select the Backup Folder you created before

Start Backup





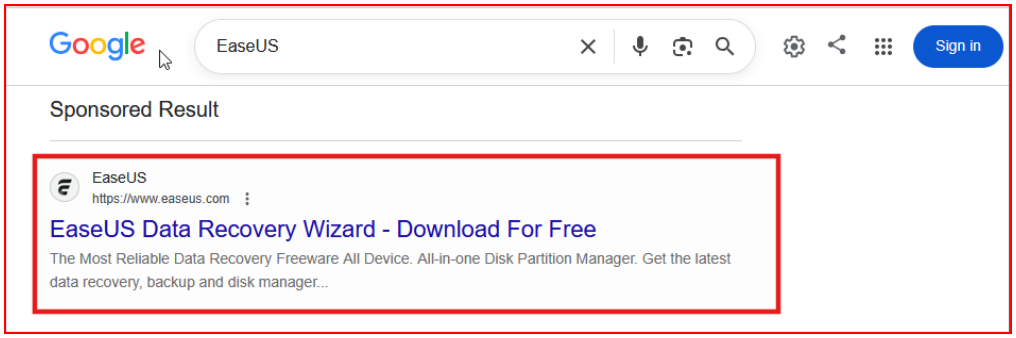
You learned how to create and run a file backup task using AOMEI Backupper, protecting selected data from accidental loss or corruption.

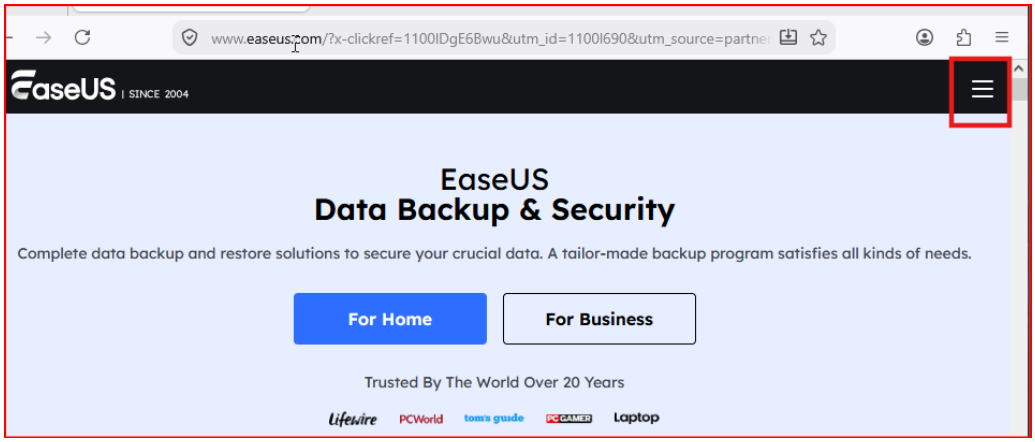
File Recovery Using EaseUS Data Recovery Wizard

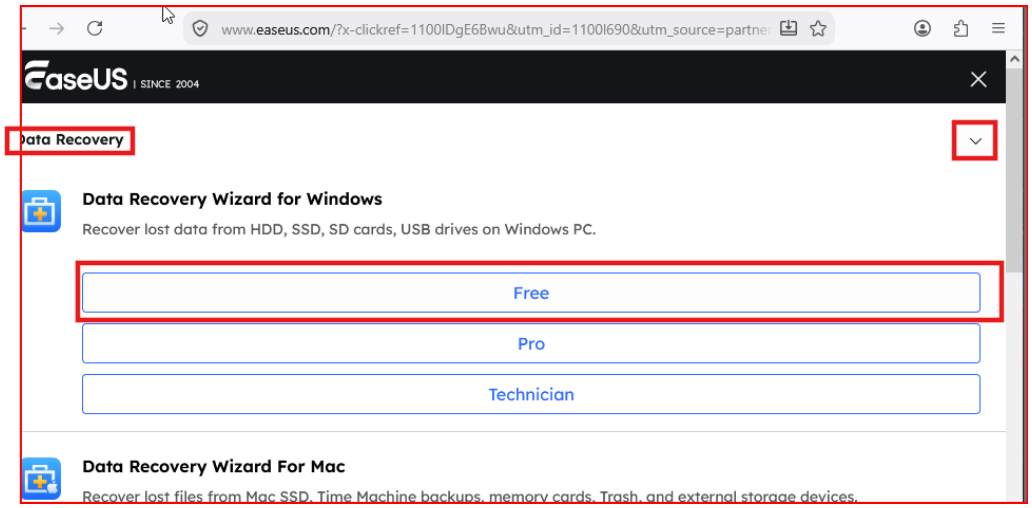
Executive Summary

This lab demonstrates data recovery after intentional deletion. On AdminMachine-1, you permanently delete the backup folder created in Lab 6, install EaseUS Data Recovery Wizard, scan Local Disk (C:), locate the deleted files, recover them to a separate folder, and verify that the recovered data opens correctly with its original size and content intact.

Installing EaseUS Data Recovery Wizard





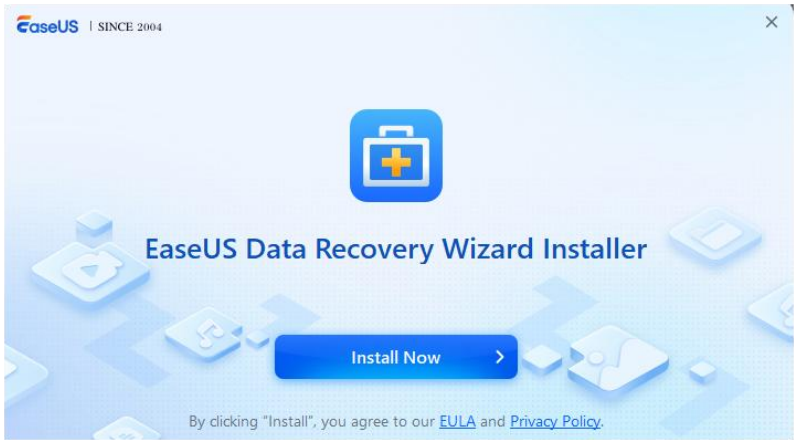


Click Free Download



Launch EaseUS Data Recovery Wizard

Click install now



Recover the Deleted Backup Folder

Preparing a Recovery Destination

1. On the AdminMachine-1 desktop, create a new folder named:

Recovered Data

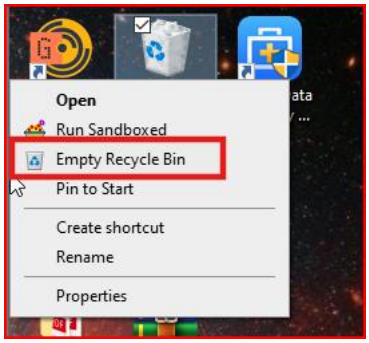


This folder will store the recovered files.

Delete the Lab 6 Backup Folder



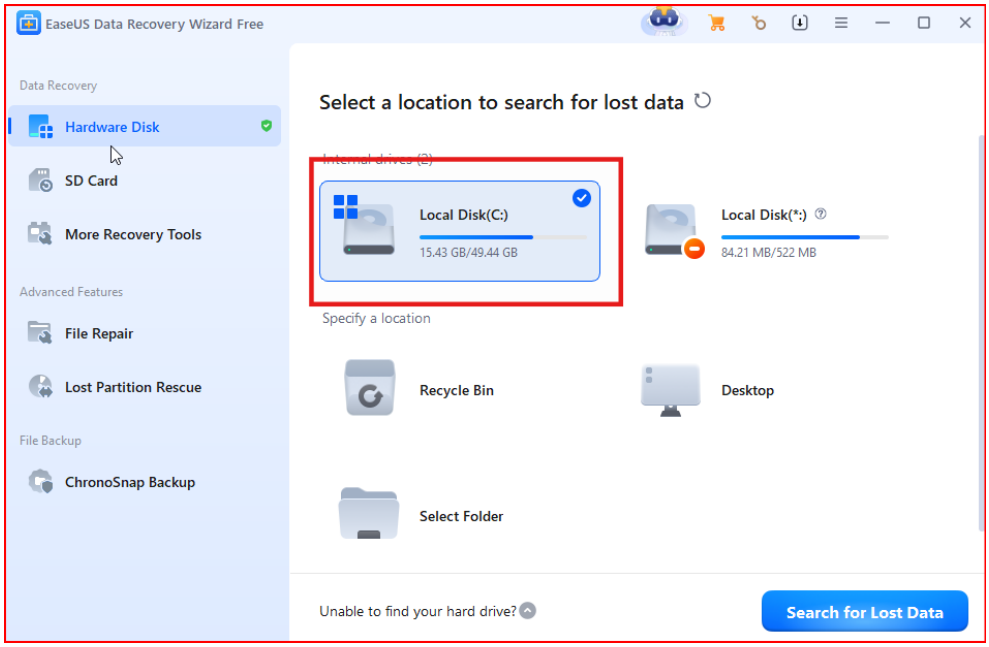
Clean the Recycle bin too



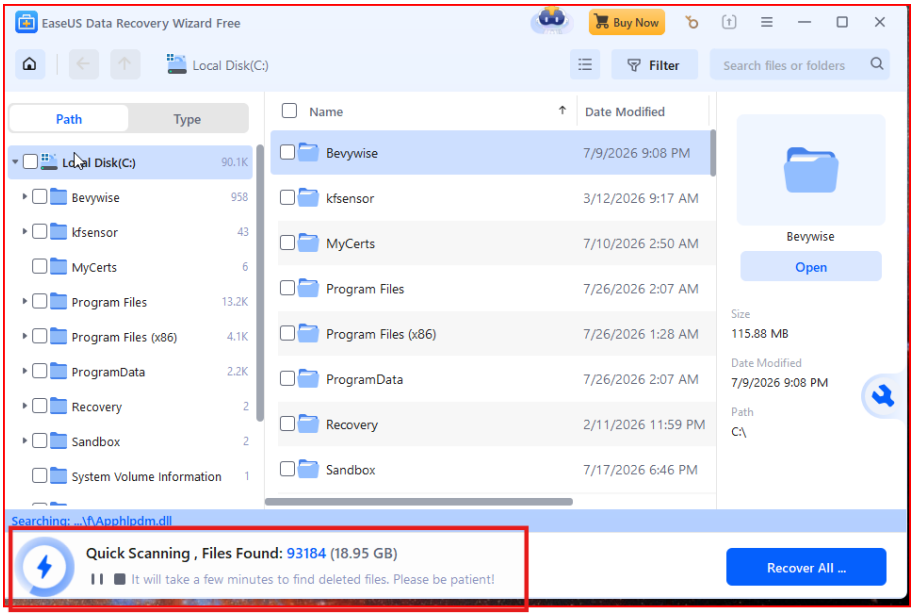
Scan Local Disk C:

1. Return to **EaseUS Data Recovery Wizard**.
2. Select:

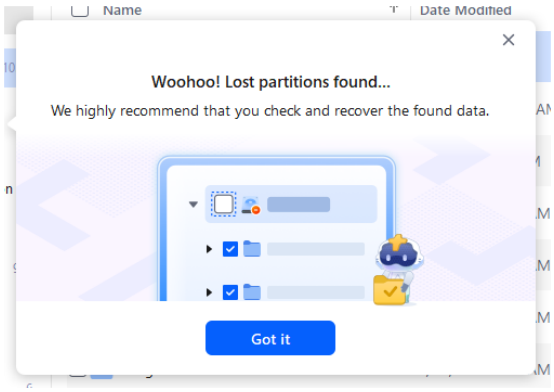
Local Disk (C:)



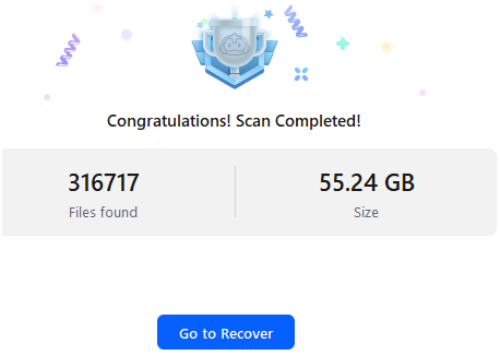
Wait till the scan completes

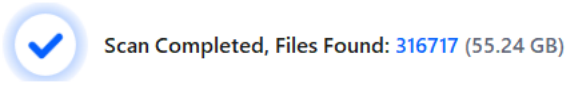


Click got it



Click go to recover





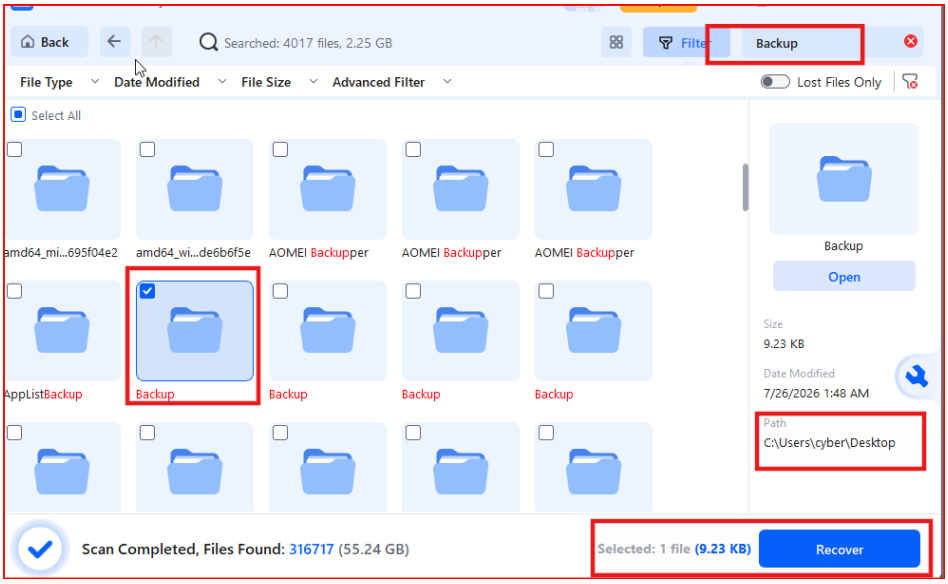
Locate the Deleted Backup Folder

1. Use the search box and enter:

Backup

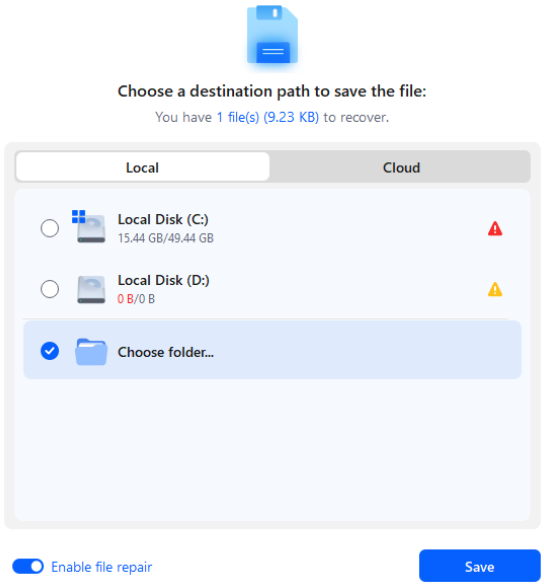
3. Look for a path similar to:

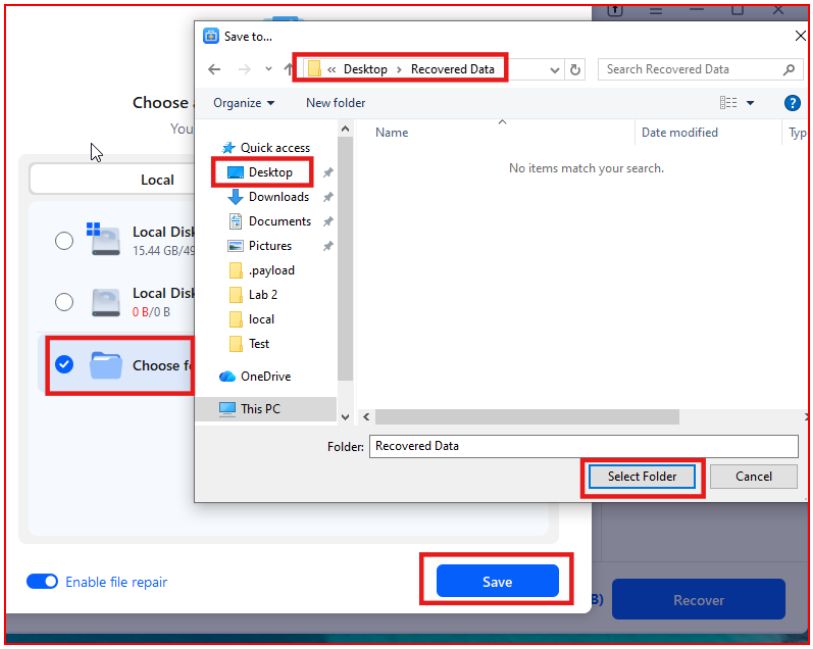
C:\Users\cyber\Desktop\Backup



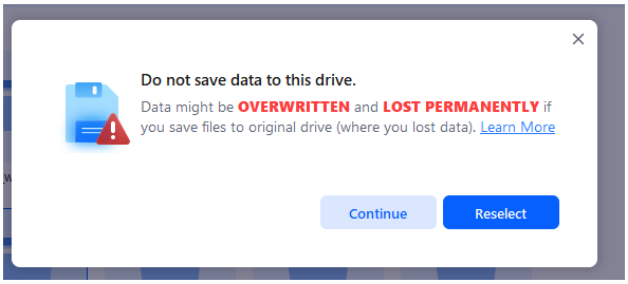
Recover the Deleted Data

Click on Choose folder and Click save





As we have a single drive C (not separated) so select continue



Verify the Recovery

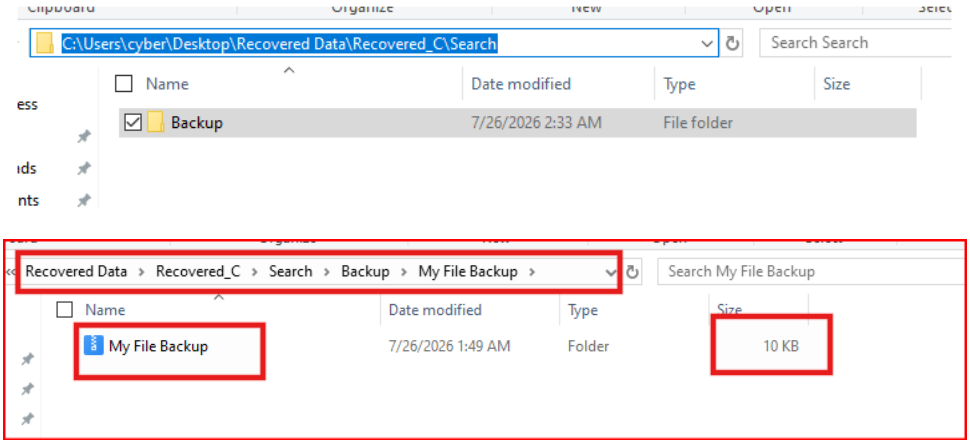
Open the recovered folders until you find the original Backup folder or AOMEI backup file.

Confirm that the recovered file:

Has its original filename

Has a file size greater than 0 KB

Opens or is visible without an error



Verification Result

- The recovered folder **My File Backup** is present.
- The recovered data size is **10 KB**.
- The recovered data is not empty or corrupted at the basic file-size verification level.

Key Learning Outcome

You learned how deleted files can be located through disk scanning and restored using data-recovery software. The recovery succeeded because the deleted disk space had not yet been overwritten.

Conclusion

All four labs in this module were completed successfully and achieved their intended objectives.

Lab 4 confirmed that SSL/TLS can effectively encrypt data in transit, securing communication between a web browser and server even when using a self-signed certificate.

Lab 5 demonstrated the practical use of OpenPGP for message confidentiality, proving that only the holder of the correct private key and passphrase can decrypt protected content.

Lab 6 showed that reliable file backups can be created quickly and easily using AOMEI Backupper, providing a clear safeguard against data loss.

Lab 7 verified that deleted files can still be recovered using specialized recovery tools, as long as the disk space has not been overwritten.

Together, these labs deliver a complete, practical understanding of core data protection techniques — encryption in transit, secure communication, backup, and recovery. The skills gained form a strong foundation for real-world cybersecurity and system administration work.